Power and Its Shadow: When Unanimity Serves the Hegemon

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Abstract

Why can unanimity benefit a powerful state when every member has the same veto? This paper studies a two-round bargaining game in which only weak states propose and a hegemon privately knows whether its disagreement payoff is low or high. Majority allows weak states to replace the hegemon's vote; unanimity makes its approval indispensable. Relative to a public-information benchmark, unanimity transfers informational rent to the low-disagreement-payoff type, while majority can eliminate that rent through exclusion. Some beliefs admit no equilibrium in the maintained pure-ballot class. An extension lets the hegemon propose first. Agenda power benefits it under unanimity, but majority can still be better when the disagreement payoff is low relative to the weak votes a winning coalition can exclude. The model offers a rationalist account of unequal bargaining power under formally equal rules.

1 Introduction

At the height of American power, the United States led the creation of the World Trade Organization (WTO). One would expect that bargaining outcomes in the trading system would reflect the asymmetry of power in the international system. Surely, International Organizations created by the hegemon would reflect such asymmetries in their rules. And yet, the WTO was created on the basis of equality rather than power: decisions by consensus meant that each member had equal veto power, and the US had no agenda power. Why would a hegemon help create an organization in which every state would formally be its equal?

The standard explanation is that equality under the law does not mean equality in practice. Coercion and information shape negotiation within institutions and can favor the powerful over the weak (Steinberg 2002). The US and the (then) European Community drafted the negotiation texts used by all, offered market access and exit options to alter other states' alternatives, assembled linked packages, and gathered information through successive proposals and objections.

The best example is the process that ended with the creation of the WTO. At the close of the Uruguay Round, they threatened states that did not accept the creation of the WTO with withdrawal from the GATT 1947. The status quo was no longer available for such states. This episode shows how material power can enter a formally egalitarian institution through the agenda, the contract space, and the value of disagreement.

From the perspective of the distributive rationalist tradition, which focuses on bargaining theories of multilateral bodies (Voeten 2019), it is hard to reconcile this explanation with a rationalist framework. Consider the standard Baron and Ferejohn framework (Baron and Ferejohn 1989). Under open rule and equal recognition of the probability of proposal, combined with unanimity, there is no asymmetry of power and, as a result, no inequality of outcomes (Kalandrakis 2006). Concretely, if the US or the so-called Quad (US, EC, Japan, and Canada) had no formal agenda power, any country could propose a deal at any stage of a bargaining round. In fact, many coalitions did that during the Uruguay Round; for example, the agro-export group of countries called the Cairns Group (Cairns Group 1987). So, either a hegemon like the US made a blunder, or there is a need to further explain the institutional architecture of institutions like the WTO from a rationalist point of view, much like Fearon (1995) did for war.

More recently, the formal literature has investigated multilateral settings in which a single uninformed player is the sole proposer and multiple symmetric responders privately know their own types. On the one hand, they highlighted how the uninformed player may optimally choose to transfer informational rents as insurance against rejection (Glynia, Thum, and Xefteris 2026). On the other hand, they show that informed responders can likewise obtain informational rents by acting tough, since rejection under unanimity can improve their continuation offers (Piazolo and Vanberg 2025).

However, their models do not study settings in which power, rather than weakness, is concentrated in a single player, a key theme in International Relations. There is also no explanation of what happens to informational rent when the same actor holds both private information and proposal power, nor of the role of screening by uninformed weak players versus signaling by the informed player when proposing. Most importantly, in these models, changing the voting rule changes how many informed voters the proposer must buy and therefore the relative appeal of pooling and screening, but every winning coalition still includes an informed responder. They leave open a more fundamental question: what happens to the informed actor's bargaining power and informational rents when a minimum winning coalition can be formed entirely from uninformed voters? This question is central to understanding the comparison between majority and unanimity from the perspective of the most powerful actor.

This paper fills this gap with a general n-player bargaining game in which power is concentrated in a single player, the hegemon, while all other players are symmetric and weak. As in the existing literature, the privately informed actor can be either a high or a low type. We first investigate what happens when even the powerful actor is stripped of proposal power. We then reintroduce proposal power by allowing the hegemon to make the first proposal, as often occurs in international organizations.

The paper's first contribution is to show that the voting rule determines whether the hegemon's private information has any bargaining value at all. Under majority, whenever the hegemon's vote is expensive relative to a weak state's, proposers assemble winning coalitions entirely from uninformed states and its informational rent is exactly zero. Under unanimity, the hegemon's approval is indispensable: proposals pool at the strong type's threshold, and the rent accrues to the hegemon whose strength is overstated. Weak states cannot distinguish the two types, so they offer enough to satisfy the high disagreement payoff. The low type accepts the same package and keeps the difference as informational rent. The baseline also has a region of beliefs with no equilibrium in the maintained class of pure ballot strategies; no institutional payoff comparison is made in that region.

The second contribution reintroduces proposal power for the informed hegemon. Moving first gives the hegemon the familiar agenda rent under unanimity, but also changes the value of its private information. Majority benefits both types when the hegemon's disagreement payoff is low relative to the share of weak states that a winning coalition can leave out. In those cases, only the informational rent can reverse the public-information advantage of majority.

Section 3 works a numerical case. Section 4 defines the game and solution concept. Section 5 solves the complete-information benchmarks, then the private games, and finally the informational rents and their difference across voting rules. Section 6 then restores proposal power to the privately informed hegemon and decomposes the effect of the agenda stage into its complete-information component and its interaction with informational rent. Section 7 interprets the pivotal-approval mechanism and its limits. The appendices give the proofs, the endpoint conventions, the exact correspondence tables, and the worked values.

2 Literature and contribution

2.1 Formal equality and informal power in international institutions

The paper begins from an established problem in international political economy. International organizations often give states formally equal rights while operating in a system with large disparities in markets, outside options, and coercive capacity (Krasner 1983; Koremenos, Lipson, and Snidal 2001). Research on informal governance shows how powerful states can preserve formally neutral rules while influencing agendas, staffing, enforcement, and the alternatives available to weaker members (Steinberg 2002; Stone 2011). The result is neither formal equality translated mechanically into equal influence nor material power operating wholly outside institutions. Power works through the bargaining environment in which formal rules are used.

Steinberg's study of the GATT and WTO supplies the closest substantive account (Steinberg 2002).

He distinguishes law-based bargaining, in which members take sovereign-equality procedures seriously, from power-based bargaining that invisibly weights outcomes toward the United States and the European Community. His historical evidence covers several practices: Green Room and Quad agenda setting, control of drafts and packages, market-access leverage, sanctions and forum exit, side payments and issue linkage, successive probes of state preferences, and legitimacy generated by an appearance of consent. These practices do not all constitute separate mechanisms. They can be organized by the strategic object each one changes.

Table 1: Steinberg's practices and their strategic counterparts

Observed practice	Strategic object	Relation to this paper
Green Room, Quad, drafting, difficult-to-amend packages	Recognition, proposal, and amendment protocol	Familiar agenda power; deliberately removed from the baseline hegemon
Market size, sanctions, alternative forums, withdrawal from GATT 1947	Disagreement and continuation payoffs	Microfoundation for asymmetric and possibly privately known $o \in \{\ell, h\}$
Side payments, exceptions, transition periods, issue linkage, single undertaking	Feasible contract space and closure rule	Motivation for a divisible package; single undertaking is distinct from destruction of the old fallback
Proposals, objections, working groups, and plenary probes	Information structure	Steinberg emphasizes information flowing from weak members to agenda setters; the model studies private information held by the hegemon
Legitimacy, implementation, and reputational concerns	Future participation and enforcement	Complementary mechanisms outside the paper's two-round baseline

This reconstruction clarifies the paper's counterfactual. The model accepts the asymmetry of disagreement values and the availability of compensating concessions, both of which Steinberg helps motivate. It then removes the hegemon's exclusive proposal power. That restriction does not assert that agenda power was absent from the WTO. It permits a cleaner question: can a formally equal approval rule give a privately informed hegemon a distributive advantage even when weak states control proposals?

2.2 Proposal rights and multilateral bargaining

Baron and Ferejohn provide the canonical framework for that question (Baron and Ferejohn 1989). Their model places recognition, proposals, voting, and continuation values inside a sequential non-cooperative game. This move makes legislative outcomes depend on the rules governing both agenda formation and approval, rather than on voting power alone. It also establishes the vocabulary used here: recognized proposers divide a fixed surplus, build winning coalitions, and compare immediate agreement with continuation.

Kalandrakis isolates the distributive force of recognition more sharply (Kalandrakis 2006). For monotonic voting rules and arbitrary discount factors, appropriate proposal probabilities can support any division of the surplus as stationary equilibrium payoffs. Proposal rights can therefore dominate the political power attributed to votes or patience. This is the benchmark the present paper switches off. The hegemon is never recognized in the baseline, so a favorable payoff cannot be attributed to its probability of proposing. Work on proposal and veto rights and the broader legislative- bargaining literature further shows that approval thresholds, continuation values, and recognition jointly determine who is bought into a coalition (Winter 1996; McCarty 2000; Eraslan and Evdokimov 2019).

Heterogeneity in disagreement values supplies the complete-information bridge. Miller, Montero, and Vanberg show that a player with an expensive known vote can benefit under unanimity and be excluded from a minimal winning coalition under majority (Miller, Montero, and Vanberg 2018). That is a public-type effect of pivotality and coalition composition. The contribution here begins after this benchmark by asking what private information adds to the same institutional comparison.

2.3 Private information and the price of approval

Two recent models identify neighboring mechanisms that the present analysis does not claim as new. Piazolo and Vanberg show how the voting rule changes the signaling value of rejection in a dynamic game (Piazolo and Vanberg 2025). Glynia, Thum, and Xefteris show how it changes an uninformed proposer's insurance problem when responders privately know their acceptance costs (Glynia, Thum, and Xefteris 2026). In both settings, every winning coalition still contains an informed responder.

The present paper instead concentrates private information in one asymmetric actor. The weak states that can substitute for the hegemon are uninformed. Majority can therefore bypass the only private information relevant to the proposed payment, whereas unanimity makes that information unavoidable.

The distinction produces two results absent from the neighboring models. First, the approval rule acts as an on-off switch for the informed participant rather than only changing the price of one informed vote relative to another. Second, the public-type benchmark permits a type-specific decomposition between the value of a necessary vote and the additional rent created by uncertainty. In the pooling region, that rent can arise through immediate agreement, without an on-path rejection. The paper therefore studies who receives the surplus when the only privately informed actor moves from replaceable to indispensable.

Related models examine majority-only screening, cheap talk, reputational toughness, informational loss, and the aggregation of signals distributed across voters (Tsai 2009; Tsai and Yang 2010; Chen and Eraslan 2013, 2014; Ma 2023; Feddersen and Pesendorfer 1998). Here weak states hold no private signals about the hegemon. The institutional comparison operates through coalition composition and the price of one privately informed actor's participation.

3 A working numerical illustration

Consider one hegemon and four weak states, with discount factor 0.9, low and high disagreement payoffs 0.10 and 0.35, and probability 0.80 of the high payoff. The terminal unanimity cutoff is

$$p^* = \frac{0.35 - 0.10}{1 - 0.10} = 0.2778.$$

The corresponding majority screening–exclusion cutoff is 0.2935. At the stated probability, private majority therefore excludes the hegemon, whereas private unanimity pools and gives both types 0.315.

The public payoff vectors under majority and unanimity are, respectively, $(0.09, 0.35)$ and $(0.09, 0.315)$. The private vectors are $(0.10, 0.35)$ and $(0.315, 0.315)$. Private information is consequently worth $(0.01, 0)$ under majority and $(0.225, 0)$ under unanimity. Consensus raises the low type's informational rent by 0.215 because weak states must pay the high threshold when they cannot replace the hegemon's vote.

For beliefs $0 < p \leq 0.2778$, the unanimity game has no equilibrium in the maintained pure-ballot class. The worked values illustrate one point at which both institutional correspondences exist; they are not an empirical calibration.

4 The model

4.1 Players, information, and proposals

There is one hegemon, H, and $m \geq 3$ weak states, $W = \{1, \dots, m\}$, for a total of $m + 1$ states. Nature selects H's terminal disagreement payoff $o \in \{\ell, h\}$, observed only by H, with $\Pr(o = h) = p \in [0, 1]$. The two possible values satisfy

$$0 < \ell < h < 1.$$

The game has two rounds. In each round, one weak state is recognized uniformly to propose; H is never a proposer. The two recognition draws are independent and made with replacement. Every weak state remains eligible in Round 2, including the Round-1 proposer. A proposal by weak state i is the allocation vector

$$x = (x_H, (x_j)_{j \in W}) \in \mathcal{X}, \qquad \mathcal{X} = \left\{ x : x_H \geq 0,\ x_j \geq 0,\ x_H + \sum_{j \in W} x_j \leq 1 \right\}.$$

Payments cannot be negative and side payments outside the proposed package are unavailable. Here x_H is the institutional concession assigned to H, and x_j is weak state j's share, including the proposer when $j = i$. The fixed unit pie is exhausted on the equilibrium path. Agreement provides H no intrinsic benefit beyond x_H, which isolates informational concessions and pivotality from any direct value of agreement.

4.2 Ballots, timing, and payoffs

The proposer is counted as voting yes. All other states vote simultaneously; the complete vote vector and outcome become public only after all votes have been cast. Define

$$k = \left\lfloor \frac{m+1}{2} \right\rfloor .$$

Under majority, passage requires k additional yes votes beyond the proposer's; unanimity requires every state's yes vote. The ballot protocol is otherwise identical across rules.

If a proposal passes, its allocations are implemented among the parties to the agreement. If H votes yes, it is a party and receives x_H. If H votes no but a majority passes the proposal without it, H is not a party: it receives its disagreement payoff o, and the share x_H is paid to no one, because the concession is specific to H and cannot be transferred to another player. Agreement and disagreement payoffs are therefore mutually exclusive at every history. In every equilibrium exclusion derived below the proposer chooses $x_H = 0$, and the hegemon receives exactly o.

If nothing passes in Round 1, no payoff is realized then and the game proceeds to Round 2. Round-2 payoffs are multiplied by $\beta \in (0, 1)$ when evaluated in Round-1 units. If nothing passes in Round 2, each weak state receives zero and H receives o in Round-2 units. A no vote is only a ballot action; it does not remove a player or trigger disagreement by itself. The delayed terminal disagreement payoff represents the cost of prolonging international negotiations.

Table 2: Transitions and payoffs in the bargaining game

Event	Weak-state payoffs	H votes yes	H votes no
Proposal passes	Proposed x_j, including x_i	x_H	the disagreement payoff o alone; x_H is paid to no one, so the total distributed falls below one when $x_H > 0$
Round-1 proposal fails	No current payoff; proceed to Round 2	$\beta C_H(\mathfrak{h}^Y)$	$\beta C_H(\mathfrak{h}^N)$
Round-2 proposal fails	Zero	o	o

Here $\mathfrak{h}^Y$ and $\mathfrak{h}^N$ denote the distinct public histories in which H voted yes or no, and $C_H(\mathfrak{h}^a)$ denotes H's Round-2 continuation value after history $\mathfrak{h}^a$. These continuation values need not be equal because the public vote can change beliefs before Round 2.

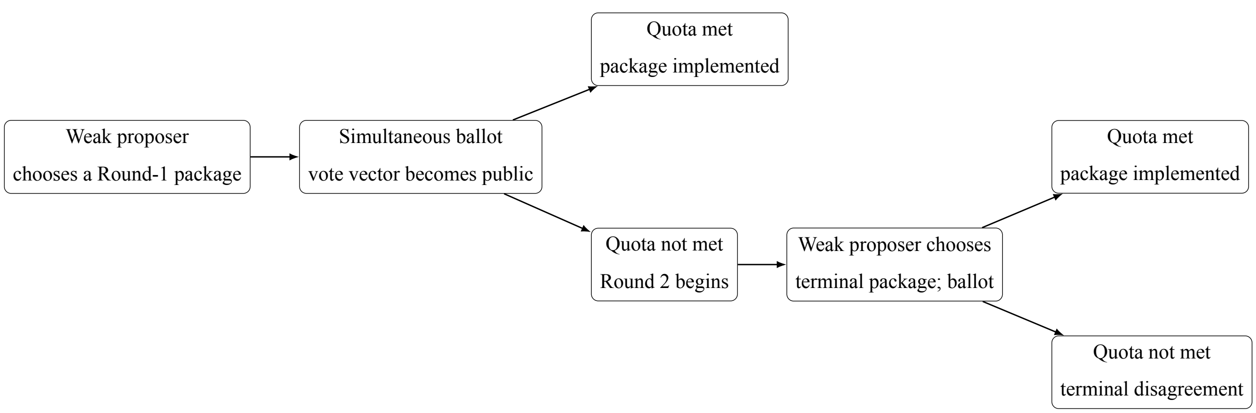

Figure 1: Sequence of proposals and ballots. Within each ballot, responding states vote simultaneously. A failed first-round proposal continues to the terminal round; a no vote alone does not end the game.

4.3 Solution concept

We use perfect Bayesian equilibrium with pure ballot strategies and three declared disciplines. First, an action by a player who does not know o does not change beliefs about o. After an action by

H, beliefs follow Bayes' rule whenever possible and otherwise satisfy structural consistency, the restriction stated exactly in Appendix A.2. Second, a weak responder evaluates yes and no as if its own vote were pivotal. Third, a voter accepts when exactly indifferent in expected value. Among proposal choices that give the proposer the same expected payoff, the selected proposal minimizes H's expected payoff.

Endpoint beliefs preserve the support of the prior. If $p = 0$, the posterior assigns zero probability to h at every history; if $p = 1$, it assigns probability one to h. For $0 < p < 1$, an action by H whose Bayesian denominator is zero under the prescribed profile and the current belief may support any posterior in $[0, 1]$. This is a condition of our solution concept. It follows the no-signaling logic used in multistage games (Fudenberg and Tirole 1991); support preservation is analogous to, but narrower than, the never-dissuaded discipline discussed by Osborne and Rubinstein (1990). Tremble consistency motivates the restriction but is not invoked as a theorem for degenerate priors (Kreps and Wilson 1982).

We use four terms consistently throughout. An *equilibrium form* describes the economic pattern—screening, pooling, exclusion, agreement, or delay. A *parameter cell* is a region of primitives and off-path beliefs on which the same characterization applies. A *complete equilibrium assessment* contains the strategies, beliefs, continuation selections, and induced outcome laws required by perfect Bayesian equilibrium. A type-payoff vector is *linked* when all of its coordinates come from the same complete assessment; for a mixed assessment, the same proposal weights therefore govern payoffs and outcomes. The model primitive o is always the terminal disagreement payoff; we reserve "outside option" for its substantive interpretation as the value of an alternative forum.

Table 3: Scope of the formal results

Element	Maintained scope
Players and horizon	One hegemon, $m \geq 3$ weak states, two rounds
Agenda	Only weak states propose in both rounds
Voting	Simultaneous public ballots; majority or unanimity
Information	Private disagreement payoff $o \in \{\ell, h\}$; support-preserving endpoints
Payoffs	Unit pie, $0 < \ell < h < 1$, $0 < \beta < 1$; no intrinsic agreement benefit for H
Equilibrium	Perfect Bayesian equilibrium in pure ballot strategies with the declared voting and proposal tie-break rules

4.4 Separate agenda-stage extension

We use superscript B for the two-round baseline and superscript A for the extension with the additional agenda stage. Thus the superscripts identify the two games being compared rather than a player or an action.

The benchmark above remains the paper's primary model. After solving it, we consider a separate extension in which the informed hegemon must make an earlier proposal at date A. Passage implements that proposal immediately; failure enters Round 1 of the benchmark and transports its complete continuation by exactly one factor β. The extension gives H no option to skip the proposal while retaining a date-A continuation. Section 6 states the economic results, and Appendices E–F preserve the complete equilibrium correspondences and date conventions.

5 Results

The solution proceeds by backward induction. We begin with public types because that benchmark separates the value of a necessary vote from the value of private information. We then solve the private terminal games, the private first-round games, and finally subtract the public benchmark component by component.

5.1 Complete-information benchmark

With a public type, no belief or signaling issue remains. The proposer compares the price of H's vote with the price of enough weak-state votes. In the terminal round, the relevant price of H is o. In the first round, it is βo, because rejection leads to the terminal round. Under majority, a weak-state vote costs β/m in first-round units. Under unanimity with public disagreement payoff o, each weak responder's continuation price is $\beta(1-o)/m$.

Proposition 5.1 (Public-type benchmark)**.** *Fix a public type with disagreement payoff o. In the terminal round, majority passes a proposal without H, assigns $x_H = 0$, and gives the weak proposer the whole unit pie; H receives o. Terminal unanimity passes with $x_H = o$, and the weak proposer retains $1 - o$.*

In Round 1, unanimity passes immediately with $x_H = \beta o$. Majority includes H if $o \leq 1/m$ and excludes it if $o > 1/m$. At equality the proposal including H is selected. Consequently, the public Round-1 payoff of H is

$$v_M^B(o) = \begin{cases} \beta o, & o \leq 1/m, \\ o, & o > 1/m, \end{cases} \qquad v_U^B(o) = \beta o.$$

Table 4: Complete-information outcomes by round and voting rule

Round	Rule	Proposed payments	Weak proposer	H
2	Majority	$x_H = 0$; weak responders get zero	1	o
2	Unanimity	$x_H = o$; weak responders get zero	$1 - o$	o
1	Majority, $o \leq 1/m$	$x_H = \beta o$; $k-1$ responders get β/m	$1 - \beta o - (k-1)\beta/m$	βo
1	Majority, $o > 1/m$	$x_H = 0$; k responders get β/m	$1 - k\beta/m$	o
1	Unanimity	$x_H = \beta o$; every responder gets $\beta(1-o)/m$	$1 - \beta o - (m-1)\beta(1-o)/m$	βo

At $o = 1/m$, buying H's vote and buying the substitute weak-state vote cost the proposer the same amount. The proposal tie-break selects inclusion because it gives H the lower payoff $\beta o < o$.

5.2 Private information in the terminal round

Let

$$p^* = \frac{h - \ell}{1 - \ell}.$$

Under terminal majority, H's vote is unnecessary and the belief never enters the proposer's choice. Under terminal unanimity, the proposer compares a low offer that succeeds only when the type is low with a high offer that both types accept.

Proposition 5.2 (Private terminal games)**.** *Under majority, the unique equilibrium outcome assigns $x_H = 0$, pays no responding weak state, and gives the full pie to the proposer. The proposal passes without H; a hegemon with disagreement payoff o receives o, and a weak state receives $1/m$ before recognition.*

Under unanimity, if $p \leq p^*$, the proposer offers $x_H = \ell$. The low disagreement-payoff type accepts and the high disagreement-payoff type rejects, so the proposal succeeds with probability $1 - p$. If

$p > p^*$, the proposer offers $x_H = h$ and both types accept. At $p = p^*$, the low offer is selected.

The cutoff equates $(1-p)(1-\ell)$, the proposer's payoff from the low offer, with $1-h$, its payoff from pooling. Notice that no discount factor appears inside this terminal comparison.

5.3 Private majority in Round 1

Let $w = \beta/m$. The proposer's payoffs from the three undominated equilibrium forms are:

$$\begin{aligned}\Pi_E &= 1 - kw,\\ \Pi_S(p) &= (1-p)\left[1-(k-1)w-\beta\ell\right] + pw,\\ \Pi_P &= 1-(k-1)w-\beta h.\end{aligned}$$

Exclusion buys k weak responders at price w each and sets $x_H = 0$. Screening buys $k-1$ weak responders and offers $\beta\ell$: the low disagreement-payoff type accepts and the high disagreement-payoff type delays. Pooling buys $k-1$ weak responders and offers βh, which both types accept. Deliberate delay is strictly worse than exclusion because $1-\beta(k+1)/m > 0$.

When $\ell < 1/m$, define

$$p_{S=E} = \frac{\beta(1/m-\ell)}{\beta(1/m-\ell)+1-\beta(k+1)/m}.$$

When $h < 1/m$, define

$$p_{S=P} = \frac{\beta(h-\ell)}{1-\beta\ell-\beta k/m}.$$

Proposition 5.3 (Private majority correspondence)**.** *For $m \geq 3$, the equilibrium form is:*

1. *If $h < 1/m$, screening for $p \leq p_{S=P}$ and pooling above it.*
2. *If $\ell < 1/m < h$, screening for $p \leq p_{S=E}$ and exclusion above it.*
3. *If $1/m < \ell < h$, exclusion for every p.*
4. *If $\ell = 1/m < h$, screening at $p = 0$ and exclusion for $p > 0$.*
5. *If $\ell < h = 1/m$, screening for $p \leq p_{S=E}$. Above that cutoff, exclusion is selected when $(1-p)\ell + ph < \beta h$, pooling is selected when the inequality is reversed, and the entire*

proposal segment joining the two survives when the two expressions are equal.

At every cutoff involving screening, screening is selected. Permuting the identities of identically paid weak responders generates additional equilibria but does not change H's payoff vector or equilibrium form. Payoffs of weak states identified by label can be permuted and, on the residual proposal family, vary with the common proposal weight.

The last segment is genuine multiplicity. Its mixing weight is a probability over pure proposals in the proposer's strategy, and the same weight governs payoffs and outcomes. The attainable set is therefore the stated one-dimensional segment.

5.4 Private unanimity in Round 1

Let

$$w_\ell^U = \frac{\beta(1-\ell)}{m}, \qquad w_h^U = \frac{\beta(1-h)}{m}.$$

At $p = 0$, support preservation prevents an impossible high type from being reintroduced by an off-path belief. The proposer can therefore offer the low threshold $\beta\ell$, pay each responding weak state its continuation w_ℓ^U, and keep $w_\ell^U + 1 - \beta$. When $p > p^*$, the continuation itself pools, so the proposer offers βh, pays each responder w_h^U, and keeps $w_h^U + 1 - \beta$.

The intermediate cell fails for a different reason. A feasible deviation can pay every weak responder enough to force a yes vote. Once all weak votes are yes, no pure contingent voting profile for the two possible types of H survives: pooling yes is rejected by the high type, pooling no is overturned by the low type at equality, and either separating profile gives one type a profitable imitation.

Proposition 5.4 (Private unanimity correspondence)**.** *Under unanimity:*

$$\begin{aligned}
p = 0 \quad &: \quad \textit{immediate low-type agreement with } x_H = \beta\ell,\ x_j = w_\ell^U,\ x_i = w_\ell^U + 1 - \beta; \\
0 < p \le p^* \quad &: \quad \textit{no perfect Bayesian equilibrium in pure ballot strategies}; \\
p^* < p \le 1 \quad &: \quad \textit{immediate pooling agreement with } x_H = \beta h,\ x_j = w_h^U,\ x_i = w_h^U + 1 - \beta.
\end{aligned}$$

The corresponding payoff vectors for H, ordered as low and high type, are $(\beta\ell, \beta h)$ at $p = 0$ and $(\beta h, \beta h)$ above p^*.

Remark (Pure-strategy scope). The middle cell establishes nonexistence only within the paper's maintained space of pure ballot strategies. We do not derive, characterize, select, or compare equilibria involving mixed ballot strategies, and make no claim about existence under such strategies.

Table 5: Private-information Round-1 equilibrium correspondences

Rule	Parameter cell	Selected equilibrium form	H payoff vector
Majority	$h < 1/m$	Screening for $p \leq p_{S=P}$; pooling above	$(\beta\ell, \beta h)$ or $(\beta h, \beta h)$
Majority	$\ell < 1/m < h$	Screening for $p \leq p_{S=E}$; exclusion above	$(\beta\ell, \beta h)$ or (ℓ, h)
Majority	$1/m < \ell$	Exclusion for every p	(ℓ, h)
Majority	$\ell = 1/m < h$	Screening at $p = 0$; exclusion for $p > 0$	$(\beta\ell, \beta h)$ or (ℓ, h)
Majority	$\ell < h = 1/m$	Screening through $p_{S=E}$; above, exclusion, pooling, or their exact proposal segment under the tie-break	$(\beta\ell, \beta h)$, (ℓ, h), $(\beta h, \beta h)$, or the linked segment
Unanimity	$p = 0$	Immediate low-type agreement	$(\beta\ell, \beta h)$
Unanimity	$0 < p \leq p^*$	No PBE in pure ballot strategies	$\emptyset$
Unanimity	$p^* < p \leq 1$	Immediate pooling agreement	$(\beta h, \beta h)$

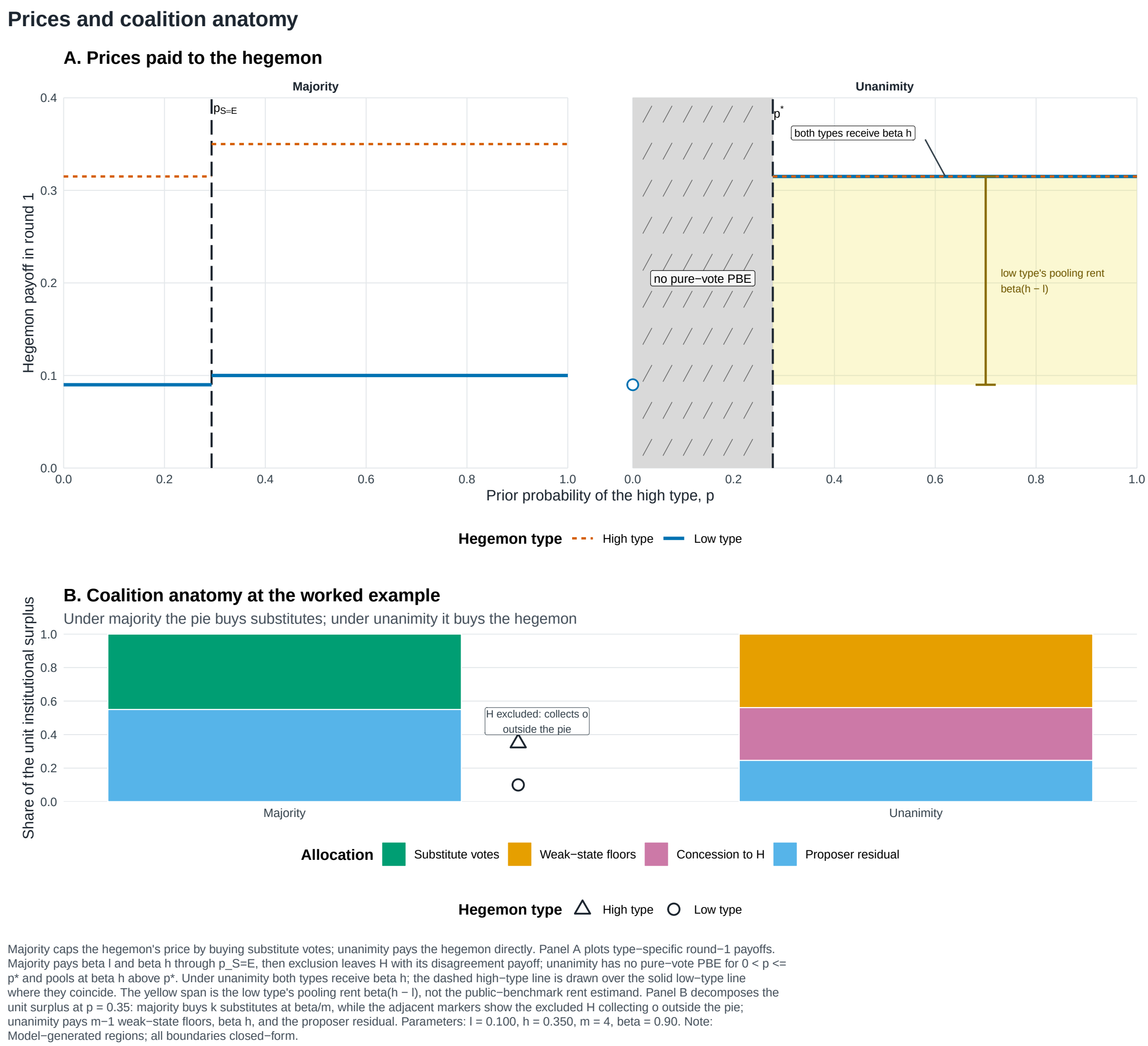

Figure 2: The price and coalition anatomy of the private games. Majority can replace the hegemon's vote with an additional weak-state vote; unanimity cannot. Labels report first-round prices and preserve the distinct low-type, pooling, and exclusion outcomes.

5.5 Comparing the private games

The comparison below uses the same parameter point under both institutions and only cells in which both games have a pure-strategy equilibrium. The timing wedges are $(1-\beta)\ell$ and $(1-\beta)h$: exclusion realizes the disagreement payoff now, whereas inclusion can price a continuation that is discounted in Round-1 units. They are distinct from the discounted type gap $\beta(h-\ell)$ and the cross-date quantity $\beta h - \ell$.

Proposition 5.5 (Private institutional payoff contrast). *Write $V_U^B - V_M^B$ as a low-type/high-type vector.*

1. *At $p = 0$, the vector is $(0, 0)$ if majority screens and $(-(1-\beta)\ell, -(1-\beta)h)$ if majority excludes.*
2. *For $0 < p \leq p^*$, the contrast is empty because private unanimity has no pure-strategy equilibrium.*
3. *For $p > p^*$, the vector is $(\beta(h-\ell), 0)$ if majority screens, $(0, 0)$ if majority pools, and $(\beta h - \ell, -(1-\beta)h)$ if majority excludes.*
4. *On the residual exclusion–pooling segment at $h = 1/m$, the exact attainable set is $\{\lambda(\beta h - \ell, -(1-\beta)h) : \lambda \in [0, 1]\}$, restricted to the proposal weights that survive the tie-break. The same λ binds payoffs and outcomes.*

Thus unanimity does not have a single unconditional payoff effect. Its effect depends on which substitute coalition disciplines the majority proposer and on which type is evaluated. Figure 3 maps these exact parameter cells without averaging types.

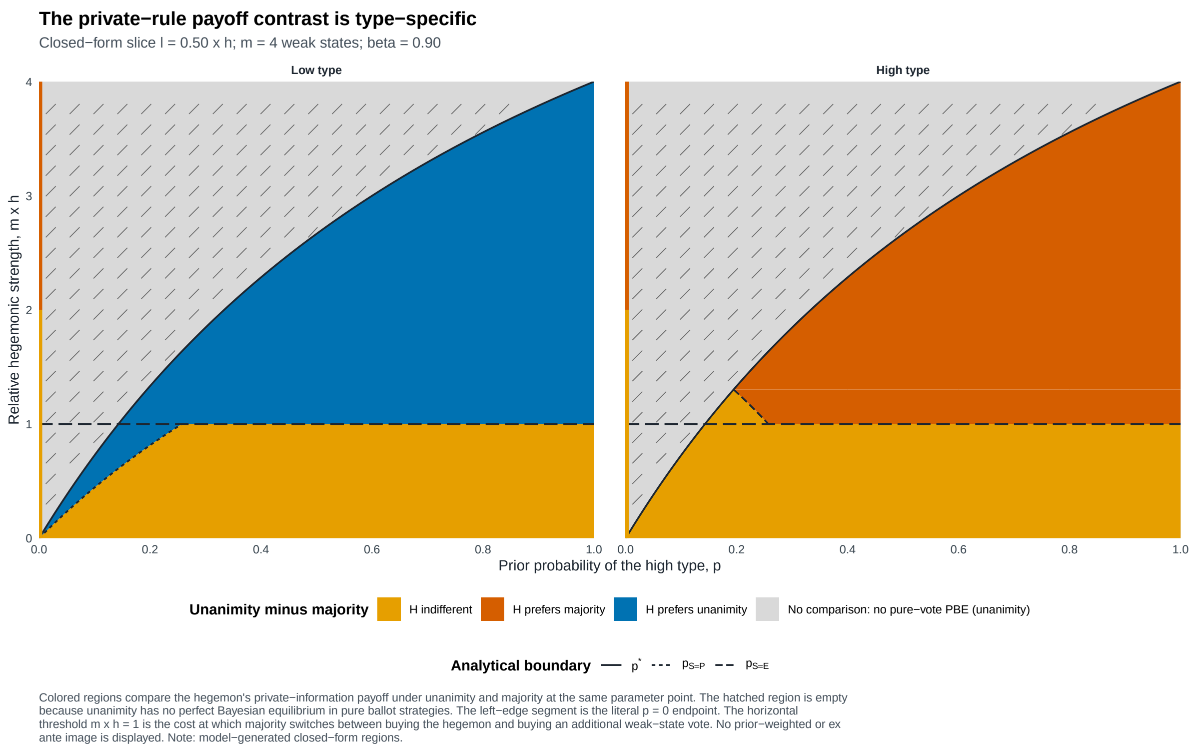

Figure 3: Private-game payoff contrast, unanimity minus majority, by hegemon type. Each cell is evaluated at the same parameter point. The hatched band marks parameter values for which private unanimity has no pure-strategy equilibrium and the contrast is therefore empty.

To read Figure 3, choose a prior on the horizontal axis and a relative high disagreement payoff mh on the vertical axis. The gray region begins below the solid boundary p^* and marks parameter cells where the unanimity comparison is unavailable. Outside it, the left panel shows when pooling under unanimity raises the low type's payoff, whereas the right panel shows when majority's ability to exclude H raises the high type's payoff. The yellow regions are payoff ties. The dotted and dashed boundaries indicate changes in the majority equilibrium form. Reading the panels at the same parameter point makes the central result visible: the same voting rule can benefit the low type and hurt the high type.

5.6 Informational rents and the difference of differences

For rule $g \in \{M, U\}$, define the type-contingent informational-rent correspondence

$$IR_g^B = V_g^B - v_g^B,$$

where the public benchmark is evaluated at the same type and the same remaining parameters. The institutional informational-rent contrast is

$$\Delta IR^B = IR_U^B - IR_M^B.$$

These are differences between correspondences. If a source correspondence is empty, so is the difference. Under multiplicity, subtraction preserves the entire attainable set and its common proposal weight.

We write the three public disagreement-payoff regions in words: both types are included when $h \leq 1/m$; the low type is included and the high type excluded when $\ell \leq 1/m < h$; and both types are excluded when $1/m < \ell$.

The rent comparison first asks what private information changes *within* each rule. Under unanimity, the public benchmark already buys the known type at its discounted threshold; private pooling therefore raises only the low type's payoff. Under majority, information matters only when the private equilibrium changes the public inclusion decision or the price paid to a type.

Proposition 5.6 (Informational rents by voting rule)**.** *Under unanimity,*

$$IR_U^B = \begin{cases} \{(0,0)\}, & p = 0, \\ \emptyset, & 0 < p \leq p^*, \\ \{(\beta(h-\ell),0)\}, & p > p^*. \end{cases}$$

Under majority, the informational rent depends jointly on the public region and the private equilibrium form:

both types included, screening	:	$(0,0)$,
both types included, pooling	:	$(\beta(h-\ell),0)$,
both types included, exclusion	:	$((1-\beta)\ell,(1-\beta)h)$,
low included, high excluded, screening	:	$(0,-(1-\beta)h)$,
low included, high excluded, exclusion	:	$((1-\beta)\ell,0)$,
both types excluded, exclusion	:	$(0,0)$.

On the residual exclusion–pooling segment at $h = 1/m$, the exact set is $\{\lambda((1-\beta)\ell,(1-\beta)h)+(1-\lambda)(\beta(h-\ell),0) : \lambda \in [0,1]\}$, restricted to the segment that survives the proposal tie-break.

In plain terms, private information creates rent only when it changes either the price paid to a type or whether that type belongs to the winning coalition. Unanimity pooling changes the low type's price; majority screening can remove that gain, while majority exclusion changes the timing of the disagreement payoff.

The difference of differences then removes majority's informational rent from unanimity's. Screening can leave a positive low-type contrast in the high-belief cell because unanimity pools. Exclusion instead compares the high threshold discounted to the present with the low current disagreement payoff, so its low-type sign is governed by $\beta h - \ell$.

Proposition 5.7 (Institutional informational-rent contrast)**.** *At $p = 0$, $\Delta IR^B = (0,0)$ when both types are included or both are excluded, while $\Delta IR^B = (0,(1-\beta)h)$ when the low type is included and the high type excluded; the high-type component there is off support but remains part of the type-contingent vector.*

For $0 < p \leq p^*$, $\Delta IR^B = \emptyset$.

For $p > p^*$, the exact vectors are

$$
\begin{array}{rcl}
\textit{both types included, screening} & : & (\beta(h-\ell),0), \\
\textit{both types included, pooling} & : & (0,0), \\
\textit{both types included, exclusion} & : & (\beta h-\ell,-(1-\beta)h), \\
\textit{low included, high excluded, screening} & : & (\beta(h-\ell),(1-\beta)h), \\
\textit{low included, high excluded, exclusion} & : & (\beta h-\ell,0), \\
\textit{both types excluded, exclusion} & : & (\beta(h-\ell),0).
\end{array}
$$

On the residual segment, the exact attainable set is $\{\lambda(\beta h-\ell,-(1-\beta)h) : \lambda \in [0,1]\}$ over the surviving proposal weights.

The institutional rent comparison therefore has a simple interpretation. When unanimity pools and majority screens, the low type keeps the discounted type gap. When majority excludes, the comparison instead weighs the high threshold paid under unanimity against the low type's current disagreement payoff. The high type moves only when private information changes its inclusion relative to the public benchmark.

In every residual-segment case, $\lambda \in [0,1]$ is restricted to the proposal weights that survive the tie-break and binds all payoff and outcome components.

The sign pattern is transparent once each component is kept separate. Above the unanimity cutoff, the low type gains $\beta(h-\ell) > 0$ when majority screens because unanimity pools and pays the high threshold. The same screening comparison is zero at $p = 0$, while the middle cell is empty. When majority excludes, the low type's gain is $\beta h - \ell$, which is positive, zero, or negative according as βh is above, equal to, or below ℓ. The high type gains $(1-\beta)h > 0$ only when private majority screens even though public majority would exclude it; it loses $(1-\beta)h$ when public majority includes it but private majority excludes. When both private rules pool, the informational-rent contrast is exactly zero.

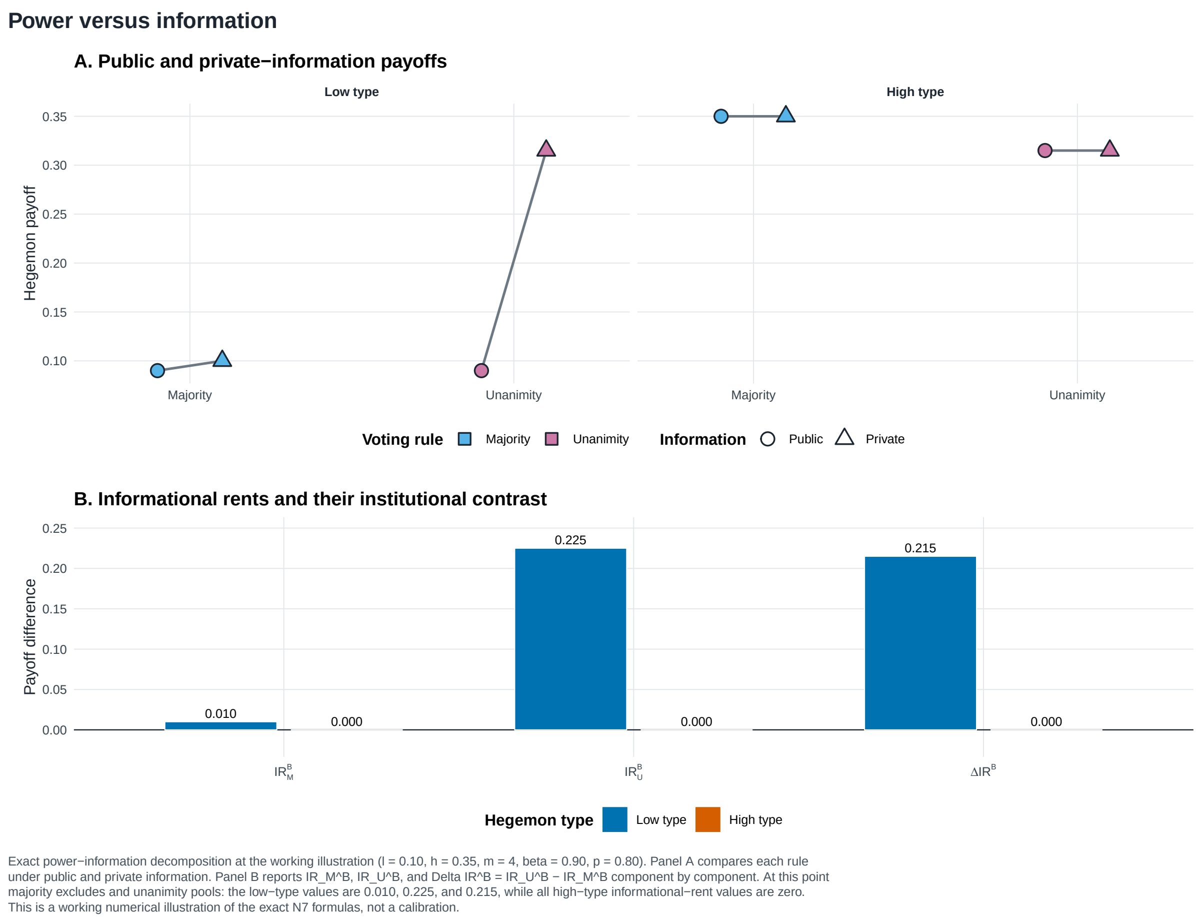

Figure 4: Power and information in the hegemon's payoff. The public benchmark captures the effect of making the hegemon's vote necessary when its type is known. Informational rent compares the private game with that benchmark, and ΔIR^B compares this rent across unanimity and majority without recombining types or selecting among multiple equilibria.

6 Agenda power and its informational shadow

The baseline removes an exclusive formal proposal privilege from H to identify informational power through pivotality. We now restore agenda power in a separate extension. Before the two-round game, at date A, the informed hegemon must propose a feasible division of the unit pie. If that proposal fails, the game enters Round 1 of the baseline. The continuation is therefore transported from Round 1 to date A by exactly one factor β. The proposal is mandatory: H cannot skip the agenda stage and obtain the continuation at date A.

Let

$$e = m - k, \qquad v_M^{\text{safe}} = 1 - \frac{k\beta}{m}.$$

Under majority, k is the number of weak votes that H must buy and e is the number of weak states that a minimal winning coalition can leave outside. The private games also retain the same off-path specification, summarized by the likelihood-ratio coordinate ρ and its implied posterior μ^{off}, where

$$\mu^{\text{off}} = \frac{p\rho}{1 - p + p\rho}$$

for an interior prior. At an endpoint, ρ is irrelevant and $\mu^{\text{off}} = p$. Comparisons across rules always pair complete equilibrium assessments from the same economy and under the same off-path specification. The linked payoff and outcome coordinates all come from the paired assessments.

6.1 Public agenda power

Suppose first that o is public before H proposes. Under unanimity, H buys every weak vote at its continuation price and agreement is immediate. Under majority, it buys a minimal coalition when passage is better than continuation, but a type with a sufficiently high disagreement payoff may deliberately induce rejection.

Proposition 6.1 (Public agenda payoffs and institutional gap)**.** *The resulting payoffs at date A are*

$$v_U^A(o) = 1 - \beta + \beta^2 o$$

and

$$v_M^A(o) = \begin{cases} 1 - \dfrac{k\beta(1 - \beta o)}{m}, & o \le 1/m, \\ \max\{v_M^{\text{safe}}, \beta o\}, & o > 1/m. \end{cases}$$

In the first majority branch, any lottery over minimal coalitions is retained. For $o > 1/m$, define

$$o_M^* = \frac{v_M^{\text{safe}}}{\beta} = \frac{1}{\beta} - \frac{k}{m} > \frac{1}{m}.$$

Agreement is immediate for $o < o_M^*$, any linked mixture of passage and delay is possible at equality, and every equilibrium delays for $o > o_M^*$. If $o_M^* \geq 1$, the last branch is outside the maintained domain $o \in (0, 1)$. Unanimity does not inherit this delay: buying all votes beats rejection by exactly $1 - \beta$.

Define the public institutional gap with the common orientation unanimity minus majority,

$$\Delta v^A(o) = v_U^A(o) - v_M^A(o).$$

Then

$$\Delta v^A(o) = \begin{cases} -\beta \dfrac{e}{m}(1 - \beta o), & o \leq 1/m, \\ \beta \left(\beta o - \dfrac{e}{m} \right), & o > 1/m \text{ and } \beta o \leq v_M^{\text{safe}}, \\ (1 - \beta)(1 - \beta o), & o > 1/m \text{ and } \beta o \geq v_M^{\text{safe}}. \end{cases}$$

The last two expressions coincide at $\beta o = v_M^{\text{safe}}$. The first branch is negative; for $o > 1/m$, the sign of $\Delta v^A(o)$ is the sign of $\beta o - e/m$. Agenda power under public information thus need not favor unanimity. If $\beta h < e/m$, majority gives both types a strictly larger payoff; equality yields a weak ranking. Above that boundary, the public component can itself point toward unanimity.

Appendices E.6–E.8 derive the public equilibrium forms and the sign characterization. The result shows that proposal power alone does not produce a uniform institutional ranking.

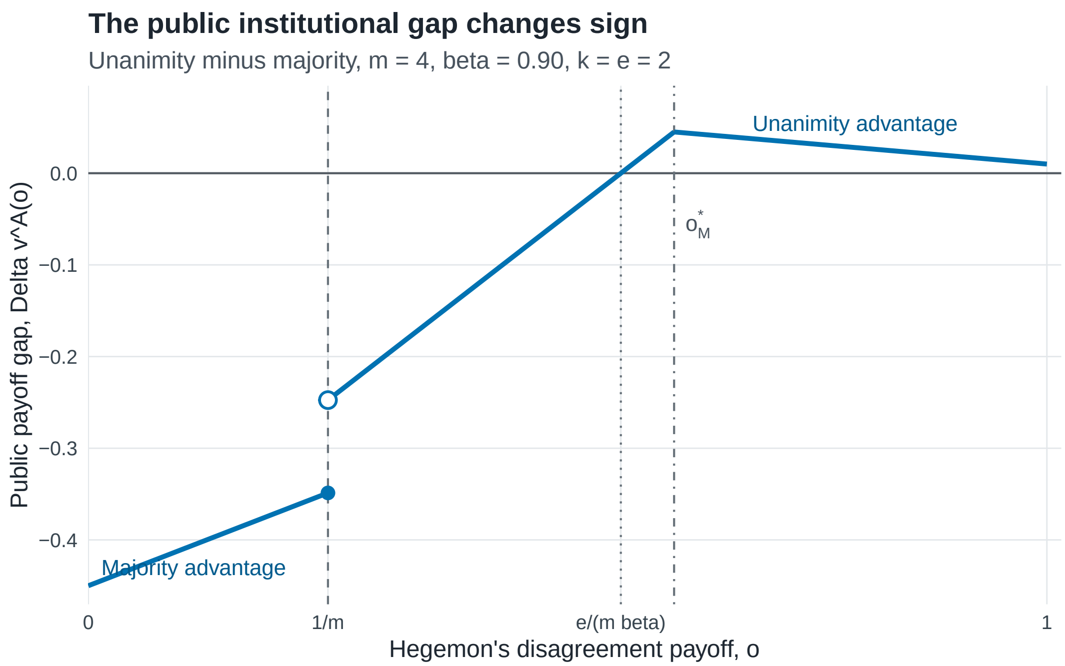

Figure 5: The public-information advantage changes with the hegemon's disagreement payoff. The curve plots $\Delta v^A(o) = v^A_U(o) - v^A_M(o)$ for $m = 4$, $\beta = .90$, and $k = e = 2$; positive values favor unanimity. Vertical reference lines mark the hegemon-inclusion threshold $1/m$, the zero crossing $e/(m\beta)$, and the majority delay cutoff o^*_M. The filled and open points at $1/m$ display the equality selection and the right-hand limit, respectively.

Figure 5 shows three regions. Below $1/m$, majority buys the hegemon's relatively cheap vote and delivers the larger public payoff. Just above $1/m$, majority substitutes weak votes for the hegemon, so its advantage persists. Once o crosses $e/(m\beta)$, the high disagreement payoff makes unanimity more valuable to H; after o^*_M, majority deliberately delays while unanimity still reaches immediate agreement.

Table 6: A worked informational reversal for the low type

Comparison	Majority	Unanimity	Unanimity minus majority
Public information	.5905	.1810	-.4095
Private information	.5905	.7290	+.1385
Informational rent	.0000	.5480	+.5480

Note: The table uses one admissible equilibrium assessment with $m = 4$, $\beta = .90$, $\ell = .10$, $h = .90$, and $p = .95$. The higher value $h = .90$, rather than the running illustration's $h = .35$, creates the region $\beta h > e/m$ in which an informational reversal is possible. The unanimity value is the lower endpoint of its admissible pooling interval. The numbers are a theoretical illustration of the decomposition.

6.2 Private agenda power and institutional comparison

Let $V_g^A = (V_g^A(\ell), V_g^A(h))$ denote the linked type-payoff vector generated by one complete private equilibrium assessment under rule $g \in \{M, U\}$. The majority correspondence contains pure pooling, separation, agreement, and deliberate-delay forms as well as Borel mixtures. All of these assessments remain in the correspondence. Under unanimity, the exact payoff correspondence can be written directly with the public value function $v_U^A(o) = 1 - \beta + \beta^2 o$:

$$V_U^A = \begin{cases} (v_U^A(\ell), \max\{v_U^A(\ell), \beta^2 h\}), & p = 0, \\ (v_U^A(\ell), v_U^A(\ell)), & 0 < p \le p^*,\ v_U^A(\ell) \ge \beta^2 h,\ \mu^{\text{off}} = 0, \\ \{(u,u) : u \in [\max\{v_U^A(\ell), \beta^2 h\}, v_U^A(h)]\}, & p^* < p < 1,\ \mu^{\text{off}} = 0, \\ (v_U^A(h), v_U^A(h)), & p^* < p < 1,\ \mu^{\text{off}} \in (p^*, 1], \\ (v_U^A(h), v_U^A(h)), & p = 1, \\ \emptyset, & \text{otherwise.} \end{cases}$$

Here and below $\emptyset$ means that the maintained perfect Bayesian equilibrium requirements have no solution in pure ballot strategies at the specified parameter and belief values.

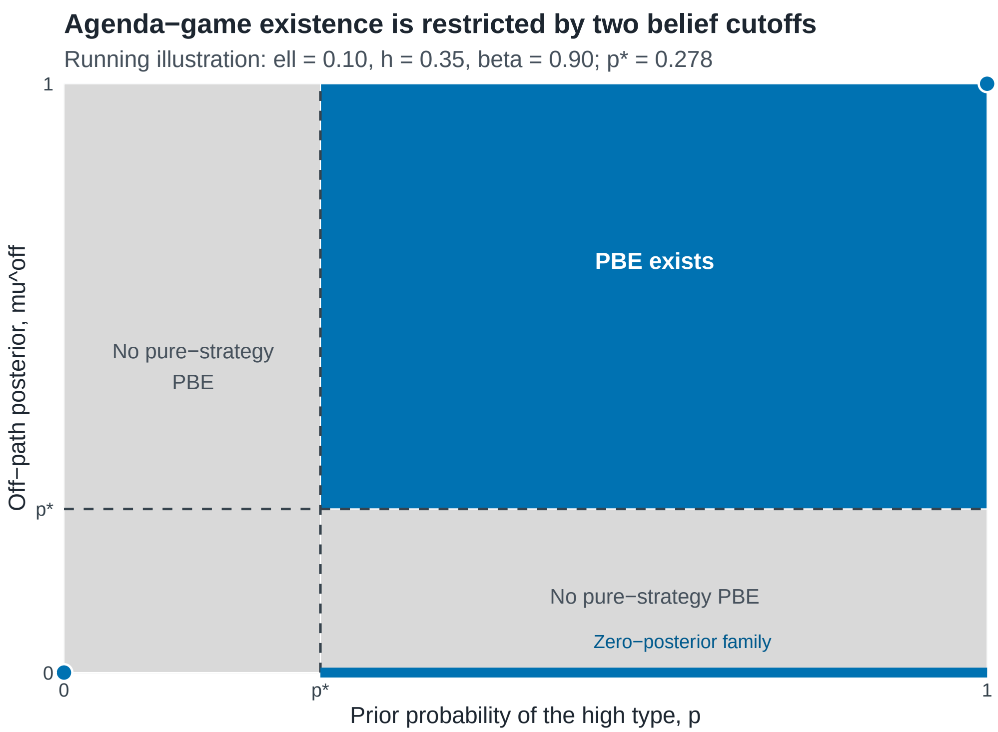

Figure 6: Existence of a pure-strategy private-unanimity equilibrium in the agenda extension. The map uses the running illustration $(m, \beta, \ell, h) = (4, .90, .10, .35)$, for which $p^* = (h-\ell)/(1-\ell) = .278$. Blue marks combinations of the prior p and off-path posterior μ^{off} with an equilibrium; gray marks nonexistence under the maintained pure-ballot concept. The blue segment at $\mu^{\text{off}} = 0$ is part of the existence set. Support preservation restricts the endpoint priors to the displayed points $(0, 0)$ and $(1, 1)$.

Figure 6 makes the domain restriction explicit. With the running values, the low-prior family fails its payoff condition, so the interior existence set begins only above p^*. In that region an off-path posterior of zero supports an interval of pooling payoffs, while posteriors above p^* select the high-posterior form. The gray regions carry no agenda-game payoff comparison.

Institutional comparison begins with the restricted product

$$\mathcal{J}_A^{\text{cmp}}(\mathbf{d}, \rho, \mu^{\text{off}}) = \mathcal{B}_M(\mathbf{d}, \rho, \mu^{\text{off}}) \times_{(\mathbf{d}, \rho, \mu^{\text{off}})} \mathcal{B}_U(\mathbf{d}, \rho, \mu^{\text{off}}).$$

For each paired assessment, define the private institutional contrast using the same unanimity-minus-majority orientation as the public contrast:

$$\Delta V^A(o) = V_U^A(o) - V_M^A(o), \qquad \Delta V_{\text{ea}}^A = (1-p)\Delta V^A(\ell) + p\Delta V^A(h).$$

Here and below, the subscript ea denotes the ex ante coordinate. The affine average is applied only after preserving the linked type vector. The comparison exists exactly when both correspondences contain an assessment at the same $(\mathbf{d}, \rho, \mu^{\text{off}})$.

The exact correspondence may remain set-valued, but one selection-free region is available.

Proposition 6.2 (Selection-free majority advantage)**.** *Fix a specification* $(\mathbf{d}, \rho, \mu^{\text{off}})$ *for which both institutional correspondences are nonempty. If*

$$\beta h < \frac{e}{m},$$

then, for both types and ex ante,

$$V_U^A(o) - V_M^A(o) \leq -\beta\left(\frac{e}{m} - \beta h\right) < 0.$$

At equality the same ranking is weak.

This condition is sufficient, not necessary. It follows from the lower bound $V_M^A(o) \geq v_M^{\text{safe}}$ and the upper bound $V_U^A(o) \leq v_U^A(h)$; outside it, the full correspondence determines the ranking. Appendices B.7–B.8 prove the two bounds. The condition is deliberately conservative: with $m = 4$, $k = e = 2$, $\beta = .9$, $\ell = .5$, $h = .6$, and $p = 0$, it fails because $\beta h = .54 > e/m = .5$, yet every majority assessment gives H more than the unanimity payoff $(.505, .505)$. Appendix E.5 supplies the complete calculation.

6.3 Public power, informational rents, and type incidence

For each complete private assessment, define the agenda-game informational rent type by type,

$$IR_g^A(o) = V_g^A(o) - v_g^A(o), \qquad \Delta IR^A(o) = IR_U^A(o) - IR_M^A(o).$$

Because every institutional contrast now uses the orientation unanimity minus majority, the decomposition is

$$\boxed{\Delta V^A(o) = \Delta v^A(o) + \Delta IR^A(o)}, \qquad \Delta IR^A(o) = \Delta V^A(o) - \Delta v^A(o).$$

The same identity holds ex ante after applying the affine map to the same linked vector.

Proposition 6.3 (Informational-rent incidence under unanimity)**.** *In every existing unanimity assessment, $IR_U^A(\ell) \geq 0$, $IR_U^A(h) \leq 0$, and at least one inequality is strict.*

Positive informational rent belongs to the low type because a pooling proposal must satisfy the high type's continuation value and the low type receives that same concession. The public benchmark already prices a known high type at its own threshold, so pooling gives that type no premium and can leave it below its public payoff. Appendix E.9 reports the exact cellwise translations, and Appendix B.8 proves the underlying exhaustive unanimity correspondence. When $\Delta v^A(\ell) < 0$, public information favors majority for the low type. If a comparable private assessment nevertheless has $\Delta V^A(\ell) > 0$, then for that assessment

$$\Delta IR^A(\ell) > -\Delta v^A(\ell) > 0.$$

Information is then the only component pointing toward unanimity and more than offsets majority's public advantage. The full correspondence retains all other admissible assessments.

Agenda also changes the informational rent relative to the no-agenda benchmark. Let IR_g^B be the Round-1 correspondence from the baseline. Its date-A value is βIR_g^B, so

$$I_g = IR_g^A - \beta IR_g^B, \qquad \Delta I = \Delta IR^A - \beta \Delta IR^B.$$

This is the only multiplication by β in the interaction. Under unanimity, agenda weakly reduces informational rent in every existing high-prior parameter cell; under majority and across rules the interaction remains set-valued wherever its sources do.

6.4 Accounting for the agenda stage

The agenda comparison places the extension with a mandatory date-A proposal and the two-round baseline on a common date scale. Under public information it compares $v_g^A(o)$ with $\beta v_g^B(o)$; under private information it compares $V_g^A(o)$ with $\beta V_g^B(o)$. Define

$$D_g(o) = v_g^A(o) - \beta v_g^B(o), \qquad T_g(o) = V_g^A(o) - \beta V_g^B(o),$$

Proposition 6.4 (Agenda-comparison decomposition)**.** *For every complete equilibrium assessment for which both comparisons exist,*

$$\boxed{T_g(o) = D_g(o) + I_g(o)}.$$

Under unanimity, T_U weakly benefits both types wherever both arms exist.

Appendix E.11 derives the identity by substituting the informational-rent definition in both games. Appendix E.13 reports the exact unanimity parameter cells. The equality is an accounting decomposition implied by $V = v + IR$ for the two model-defined comparisons. Under multiplicity it is a correspondence of admissible differences.

The public effects are

$$D_U(o) = 1 - \beta$$

and

$$D_M(o) = \begin{cases} v_M^{\text{safe}} - \dfrac{e}{m}\beta^2 o, & o \le 1/m, \\ v_M^{\text{safe}} - \beta o, & 1/m < o < o_M^*, \\ 0, & o \ge o_M^*. \end{cases}$$

Thus agenda never lowers H's public payoff under majority, but it can have no direct value when the mandatory proposal induces the same delayed continuation as the baseline. The first branch applies at $o = 1/m$; the specified continuation selection creates a downward jump in D_M immediately to its right. The threshold o_M^* can lie outside $o \in (0, 1)$.

Under private majority,

$$T_M = D_M + I_M$$

is the exact translation of the interaction correspondence by the fixed public vector. Its sign can vary across the correspondence. Across rules, assessment by assessment within a complete admissible product,

$$\Delta D(o) = D_U(o) - D_M(o), \qquad \Delta T(o) = T_U(o) - T_M(o) = \Delta D(o) + \Delta I(o).$$

The diagonal object $Q_g = v_g^A - \beta V_g^B$ changes agenda and information simultaneously and therefore reports a composite model comparison; Appendix E presents it separately.

7 Discussion

7.1 Pivotality, substitutes, and contested strength

The model's central distinction is technological. Under majority, the proposer can replace the hegemon with another uninformed weak-state vote. Under unanimity, the hegemon's approval is an essential input. Removing the substitute changes both bargaining power and the way private information enters proposals.

The complete-information benchmark shows the power component. If the known hegemon is cheap enough, majority includes it and both rules pay the same discounted threshold. If it is expensive, majority excludes it while unanimity must still buy its vote. The informational comparison asks a different question: after netting out that public-type difference, how much does private information change each rule's payoff?

The answer is not simply that private information always benefits a hegemon. Pooling under unanimity benefits the low type because weak states price it as high. Screening under majority can remove that rent, while exclusion can make the comparison depend on whether the high threshold discounted to the present exceeds the low current disagreement payoff. The high type's rent changes only when its inclusion or price differs across the private and public majority games.

The empty middle-belief cell is equally important for scope. It is the domain in which every pure voting pattern fails one of the declared belief or pivotality conditions. Figure 3 displays this domain as the gray hatched band, which carries no institutional payoff comparison.

7.2 WTO creation, exit, and observable implications

The close of the Uruguay Round provides a sharp institutional illustration. Steinberg reports that the United States and the European Community backed the single undertaking and, after joining the WTO, withdrew from GATT 1947 (Steinberg 2002). Their withdrawal terminated the old GATT obligations, including most-favored-nation treatment, toward countries that did not accept the Final Act and join the new organization. The maneuver made the preexisting trading relationship a less valuable fallback and helped give the WTO package broad membership.

The two parts of that strategy map to different bargaining objects. The single undertaking joined the Uruguay Round agreements into one package and restricted selective acceptance. Withdrawal from GATT 1947 changed the value of rejecting that package. The latter supplies a concrete microfoundation for disagreement payoffs: market power permits a major state to shape what agreement failure is worth to itself and to its partners. If the precise value of the major state's outside option is not common knowledge, bargaining over an approval that cannot be replaced also creates the screening problem analyzed in the model.

The episode is not a literal implementation of the baseline. Steinberg shows that the United States and European Community exercised substantial de facto agenda influence, whereas the benchmark denies the hegemon an exclusive formal proposal right. The benchmark isolates one channel inside a historically richer process. It asks what the consensus requirement contributes after agenda control is held fixed, not whether agenda control was absent or whether the model identifies why the United States created the WTO.

This distinction yields observable implications. Informational power through pivotality should be strongest where a major state's approval has no coalition substitute, its fallback is hard for partners to price, and the negotiated package can vary enough to compensate uncertain participation constraints. In the pooling region, the signature is a smooth agreement that grants a concession suited to a stronger possible type. A theory of signaling through rejection instead predicts visible delay or failed proposals before beliefs change. Under majority, the availability of a substitute coalition should reduce the return on private information even when the underlying outside option remains unequal.

Steinberg also emphasizes information flowing in the opposite direction: the consensus process helps powerful agenda setters elicit weaker states' preferences. The model studies weak proposers who are uncertain about the hegemon's continuation value. An empirical application must distinguish these directions of information and use language of consistency rather than treat the WTO episode as identification of the formal mechanism.

7.3 Limits

The baseline fixes the voting rule, keeps the institutional pie at one, gives proposal power only to weak states, and uses two types and two rounds. The agenda extension relaxes only that baseline proposal restriction by adding the earlier mandatory proposal stage for H. Neither part analyzes endogenous rule choice or an endogenous agenda protocol; the paper also does not analyze more than two types, sequential roll-call voting, or intrinsic agreement benefits. The restriction $m \geq 3$ excludes the three-player case. The strict discounting assumption excludes the corner $\beta = 1$, where additional indifferences would require a separate analysis.

The endpoint-support condition also matters. A type assigned zero probability by the prior cannot be resurrected after an off-path action. This makes the degenerate-prior cells coincide with their complete-information games. In the interior, both types remain in the prior's support and off-path actions by H can sustain a range of posterior beliefs.

The agenda extension preserves rather than resolves equilibrium multiplicity. Its majority effects, institutional comparisons, and some interactions are correspondences. Every robust sign is therefore setwise, and every other sign is assessment-specific. Empty cells contain no pure-strategy PBE under the maintained solution concept. The agenda comparison T depends on the maintained date convention and mandatory proposal stage; the diagonal comparison Q changes both the agenda and information specification.

Finally, the WTO discussion is a theoretical application. The model neither identifies why the United States and European Community designed the WTO nor establishes that a particular historical actor was a low or high type. It also abstracts from plurilateral agreements, subsidiary voting procedures, ratification, implementation, and repeated-game reputation. Those mechanisms may complement or substitute for the channels isolated here.

8 Conclusion

Consensus can benefit a hegemon even when weak states control the agenda because unanimity removes substitutes for the hegemon's vote. Complete information reveals the familiar pivotality component. Private information adds a distinct component: when unanimity pools at the high threshold, a low type can earn rent; whether this exceeds its majority rent depends on the majority correspondence and the disagreement-payoff region.

Restoring hegemonic agenda power does not erase the benchmark result. It separates real power from its informational shadow. When the disagreement payoff is high, the public component can itself favor unanimity. When public information favors majority, a private unanimity ranking can arise only if the relative informational rent offsets that public disadvantage. The agenda-stage accounting identity is $T = D + I$: its public-information component and its interaction with information are distinct, and neither has a universal institutional sign.

Formal equality of votes can therefore coexist with unequal bargaining power through three channels: outside options, proposal rights, and the price of an approval with no substitute. The WTO application is consistent with their coexistence, while the identifying benchmark shows that the last channel does not require an exclusive formal proposal right. The paper's conclusions remain conditional on complete equilibrium correspondences, their domains, and the cells in which the maintained equilibrium concept exists.

Appendix A: Protocol, beliefs, and incentives

A.1 Complete transition and payoff rules

For a proposal $x = (x_H, x_1, \dots, x_m)$, let $a_H \in \{Y, N\}$ and let the weak responders' votes be a_{-i}. The proposer votes yes. If the quota is met, the proposal is implemented among the parties. Each weak respondent receives x_j, the proposer receives x_i, and H receives x_H after yes; after no, H receives o and x_H is paid to no one. If the quota is not met in Round 1, the public vote vector leads to Round 2; every continuation payoff is then multiplied by β. If the quota is not met in Round 2, weak states receive zero and H receives o.

Agreement and disagreement payoffs are mutually exclusive at every history: no history pays a player both an allocation from the pie and a disagreement payoff. All equilibrium exclusions pay H exactly o.

A.2 Beliefs and ballot restrictions

Uninformed weak-state proposals and votes preserve the current belief. Structural consistency is the following restriction. Fix a profile. At each ballot, the entering belief is the current posterior, and H's prescribed law is the type-contingent vote distribution the profile assigns at that ballot. The posterior after any vote vector depends on the history only through the entering belief, the prescribed law, and H's realized vote; it is therefore invariant across vote vectors of the same ballot that differ only in weak-state votes. If H's realized vote has positive probability under the prescribed law and the entering belief, the posterior is the Bayes update given that law, including at ballots reached by earlier weak-state deviations. If the Bayes denominator is zero, the posterior is a single free value attached to that ballot-and-vote pair, chosen within the support of the prior: any value in $[0, 1]$ for an interior prior, and the degenerate posterior at $p = 0$ or $p = 1$. Distinct ballots may carry distinct free values. This condition is a declared restriction of our equilibrium concept; it is not the consistency notion of sequential equilibrium (Kreps and Wilson 1982), and it is narrower than the never-dissuaded discipline of Osborne and Rubinstein (1990).

A weak responder compares its allocation with the continuation that would follow if its vote switched passage into failure. It votes yes at equality. The hegemon maximizes its type-contingent continuation payoff and votes yes at exact indifference. These restrictions apply at every proposal, including off-path proposals.

Appendix B: Proofs

B.1 Proof of Proposition 5.1

In terminal majority, a weak responder's disagreement value is zero, so any nonnegative allocation induces yes under the indifference-to-yes convention. There are $m-1$ weak responders, and $k = \lfloor (m+1)/2 \rfloor \leq m-1$ for $m \geq 3$; their votes can therefore pass the proposal without H. At any such proposal, a nonpivotal H votes yes exactly when $x_H \geq o$: yes yields x_H, whereas no yields o. Yet every $x_H > 0$ is strictly suboptimal for the proposer. If H votes no, x_H is paid to no one; if H votes yes, it is paid for a vote that is unnecessary for passage. In either case, setting $x_H = 0$ and assigning that amount to the proposer preserves every weak responder's allocation and ballot, preserves passage, and raises the proposer's payoff by x_H. Positive payments to weak responders can likewise be reduced to zero without changing their ballots. Hence the unique equilibrium outcome has $x_H = 0$, zero payments to every weak responder, the proposer keeps the unit pie, and H votes no and receives o.

In terminal unanimity, every weak responder accepts zero and H accepts exactly when $x_H \geq o$. The proposer therefore sets $x_H = o$.

Move to Round 1. The terminal continuation of a weak state before recognition under majority is $1/m$, hence $w = \beta/m$. The continuation of H is βo. Under unanimity, paying H exactly βo and each weak responder its continuation implements immediate agreement and leaves the proposer strictly more than delay because $\beta < 1$.

Under majority, inclusion costs $(k-1)w + \beta o$; exclusion costs kw. Inclusion is weakly cheaper exactly when $o \leq 1/m$. At equality the proposer is indifferent, and the proposal tie-break selects inclusion because H receives $\beta o < o$. This gives the stated payoffs. $\square$

B.2 Proof of Proposition 5.2

The majority argument is the terminal-majority part of the preceding proof. Before recognition, symmetry gives each weak state expected payoff $1/m$.

Under unanimity, weak responders accept zero. Any accepted offer to H can be reduced to the lowest threshold consistent with its acceptance set. The only nonempty acceptance sets are the low type alone, implemented by $x_H = \ell$, and both types, implemented by $x_H = h$. The first gives the proposer $(1-p)(1-\ell)$; the second gives $1-h$. The low offer is weakly preferred exactly when

$$p \leq \frac{h-\ell}{1-\ell} = p^*.$$

At equality the low offer gives H the lower expected payoff and is selected. □

B.3 Proof of Proposition 5.3

A responding weak state votes yes exactly when $x_j \geq w = \beta/m$. This threshold is independent of the posterior and of H's vote because a weak state's terminal-majority continuation is always $1/m$. Let n_Y be the number of such responders. If $n_Y \geq k$, the weak-state votes pass the proposal without H. A nonpivotal type with disagreement payoff o then votes yes exactly when $x_H \geq o$: yes yields x_H, whereas no yields o. Nevertheless, every $x_H > 0$ in this case is strictly suboptimal for the proposer. Moving x_H to the proposer's own allocation preserves the weak-state votes and passage and raises the proposer's payoff by x_H. Hence the optimal exclusion candidate sets $x_H = 0$; both types then vote no and receive their disagreement payoffs. If $n_Y = k-1$, H is pivotal and a type with disagreement payoff o votes yes exactly when $x_H \geq \beta o$, because no leads to the terminal-majority continuation. If $n_Y \leq k-2$, the proposal fails regardless of H's vote; the two ballot actions deliver the same terminal-majority continuation, so the indifference-to-yes convention selects yes.

The preceding dominance argument leaves only $x_H = 0$ in the class $n_Y \geq k$. In the pivotal class $n_Y = k-1$, weak payments can be reduced to w and x_H to $\beta\ell$ or βh, according to the desired acceptance set; proposals in the remaining class deliver delay. Assigning the remaining pie to the proposer weakly increases its payoff. This leaves four candidates: exclusion with payoff

Π_E, screening with payoff $\Pi_S(p)$, pooling with payoff Π_P, and deliberate delay with payoff w. Exclusion is always feasible and

$$\Pi_E - w = 1 - \frac{\beta(k+1)}{m} > 0,$$

so delay is never selected. A screening or pooling candidate that could beat exclusion is automatically feasible.

The decisive payoff differences are

$$\Pi_P - \Pi_E = \beta(1/m - h)$$

and

$$\Pi_S(p) - \Pi_E = (1-p)\beta(1/m - \ell) - p(1 - \beta(k+1)/m).$$

Where pooling can beat exclusion, comparing screening with pooling yields $p_{S=P}$; comparing screening with exclusion yields $p_{S=E}$. Substituting the signs of $\ell - 1/m$ and $h - 1/m$ produces the five cases in Proposition 5.3.

At screening ties, its expected payoff to H is strictly smaller than that of the tied alternative, so the proposal tie-break selects screening. When $h = 1/m$, exclusion and pooling give the proposer the same payoff. Their payoffs to H determine the stated selection; exact equality leaves their entire proposal segment. All selected proposals exhaust the pie. Finally, permuting the identities of weak responders does not change any comparison, which establishes the reported identity multiplicity. □

B.4 Proof of Proposition 5.4

Transport the terminal unanimity continuation into Round-1 units. If the posterior is μ, a weak state's continuation is

$$c_U(\mu) = \begin{cases} (1-\mu) w_\ell^U, & \mu \le p^*, \\ w_h^U, & \mu > p^*. \end{cases}$$

Thus $c_U(\mu) \in [w_h^U, w_\ell^U]$, and $x_j = w_\ell^U$ forces every weak responder to vote yes under any admissible posterior.

At $p = 0$, support preservation fixes every posterior at zero. The proposal

$$x_H = \beta\ell, \qquad x_j = w_\ell^U, \qquad x_i = 1 - \beta\ell - (m-1)w_\ell^U = w_\ell^U + 1 - \beta$$

is accepted. Each responder and H is exactly indifferent and votes yes. Any lower payment loses a required vote; any higher payment reduces the proposer's residual. A delay gives the proposer w_ℓ^U, so immediate agreement is strictly better by $1 - \beta$.

For $p > p^*$, the analogous pooling proposal is

$$x_H = \beta h, \qquad x_j = w_h^U, \qquad x_i = 1 - \beta h - (m-1)w_h^U = w_h^U + 1 - \beta.$$

Both types of H and every weak responder accept, and immediate agreement again exceeds delay by $1 - \beta$.

For completeness, write $u = \min_j x_j$ after an arbitrary proposal and order the strategy of H as low-type/high-type. At $p = 0$, posterior support is always the low type. The profile (Y, Y) is sequentially rational if $u < w_\ell^U$, or if $u \geq w_\ell^U$ and $x_H \geq \beta h$; (N, N) is rational if $u \geq w_\ell^U$ and $x_H < \beta\ell$; and (Y, N) is rational if $u \geq w_\ell^U$ and $\beta\ell \leq x_H < \beta h$. The profile (N, Y) is never rational. These cases are mutually exclusive and exhaustive.

For $p^* < p < 1$, the profile (Y, Y) is rational if $u < w_h^U$, or if $u \geq w_h^U$ and $x_H \geq \beta h$; (N, N) is rational if $u \geq w_h^U$ and $x_H < \beta h$; and neither separating profile is rational. Bayes gives the current prior after an on-profile pooled action. When yes is a zero-probability action, an admissible posterior μ_Y supports all weak yes votes exactly when $c_U(\mu_Y) \leq u$, which is possible in the stated no profile because $u \geq w_h^U$. At $p = 1$, support preservation fixes the posterior at one and the same two pooled-profile conditions apply. Thus every proposal in both existence domains has a pure, sequentially rational response completion.

It remains to show the empty cell. For $0 < p \leq p^*$, consider

$$s^\dagger: \quad x_H = \beta\ell, \quad x_j = w_\ell^U \text{ for every weak responder}, \quad x_i = 1 - \beta\ell - (m-1)w_\ell^U.$$

All weak responders must vote yes. Enumerate the four pure type-contingent profiles for H, ordered low/high:

1. Under (Y, Y), the high type receives $\beta\ell < \beta h$ and profitably votes no to obtain continuation βh.
2. Under (N, N), the low type's no gives $\beta\ell$, so the indifference-to-yes convention requires yes.
3. Under (Y, N), the low type can imitate the high type's no. That action reveals the high type and yields continuation $\beta h > \beta\ell$.
4. Under (N, Y), the high type can imitate the low type's no and obtain continuation βh, while the prescribed comparison cannot support strict no against yes at the offer.

No pure profile is sequentially rational after $s^{\dagger}$. Because a PBE must specify a valid pure response after every feasible proposal, no pure-ballot PBE exists in this cell. This also explains the discontinuity: at $p = 0$, support preservation rules out the posterior that drives the imitation argument; for every positive p, both types are possible. □

B.5 Proof of Proposition 5.5

The private majority payoff vectors for H are

$$V_M^{B,S} = (\beta\ell, \beta h), \quad V_M^{B,P} = (\beta h, \beta h), \quad V_M^{B,E} = (\ell, h).$$

The private unanimity vectors are $(\beta\ell, \beta h)$ at $p = 0$, empty for $0 < p \leq p^*$, and $(\beta h, \beta h)$ for $p > p^*$. Componentwise subtraction gives the proposition. On any proposal segment, subtraction is affine in the common proposal weight, so the payoff set and outcome set remain atomic. □

B.6 Proofs of Propositions 5.6 and 5.7

From Proposition 5.1, the public payoff vectors are

$$v_M^B = (v_M^B(\ell), v_M^B(h)), \qquad v_U^B = (\beta\ell, \beta h).$$

Subtracting these from each private vector gives the table in Proposition 5.6. In particular, unanimity is payoff-equivalent to its public endpoint at $p = 0$, is empty in the middle cell, and pools above the cutoff, which yields $(\beta(h-\ell), 0)$.

Subtract IR_M^B from IR_U^B. In the high-belief cell, screening gives $(\beta(h-\ell), 0)$ when both types are publicly included and $(\beta(h-\ell), (1-\beta)h)$ when only the low type is publicly included; pooling gives zero. Exclusion gives $(\beta h-\ell, -(1-\beta)h)$, $(\beta h-\ell, 0)$, or $(\beta(h-\ell), 0)$, respectively when both types are included, only the low type is included, or both types are excluded. The endpoint follows by the same calculation. An empty source set remains empty. For the residual segment, affine subtraction with its common weight λ yields the one-dimensional segment stated in the propositions. □

B.7 Proof of the private-majority agenda correspondence

Fix the primitives. Let

$$\mathfrak{C}_M = \{\mathsf{E}, \mathsf{S}, \mathsf{P}\} \sqcup (\{\mathsf{EP}\} \times [0,1])$$

be the compact space of admissible majority-continuation states and fix an anonymous Borel Markov selection $\chi : [0,1] \to \mathfrak{C}_M$. For posterior μ, let $c_\chi(\mu)$ be a weak state's continuation payoff and $h_{\chi,o}(\mu)$ the continuation payoff of type o, both in the native units of the baseline. Transporting the continuation to date A gives

$$r_\chi(\mu) = \beta c_\chi(\mu), \qquad d_{\chi,o}(\mu) = \beta h_{\chi,o}(\mu).$$

Every baseline branch gives $0 < c_\chi(\mu) \leq 1/m$. Hence

$$0 < r_\chi(\mu) \leq \frac{\beta}{m}, \qquad kr_\chi(\mu) < 1.$$

Under as-if-pivotal voting, weak state j votes yes if and only if $x_j \geq r_\chi(\mu)$; equality passes. The optimal passing proposal buys exactly k votes at this price. Hence H's retained share is

$$a_\chi^{\text{pass}}(\mu) = 1 - kr_\chi(\mu).$$

A proposal such as $(1, 0, \dots, 0)$ is rejected and attains $d_{\chi,o}(\mu)$. Thus both terms in

$$\max\{a_\chi^{\text{pass}}(\mu), d_{\chi,o}(\mu)\}$$

are attainable at every fixed posterior.

For a pure interior profile, the public support contains at most two proposals. Its complement is dense in the proposal simplex. If the canonical best passing proposal at μ^{off} is itself on the support, reduce H's share by $\varepsilon > 0$, keep the k threshold payments fixed, and choose a sequence $\varepsilon \downarrow 0$ that avoids the finitely many supported proposals. Rejected proposals are also available off support. Consequently, the supremum payoff from an unsupported proposal is exactly

$$O_o(\rho) = \max\{a_\chi^{\text{pass}}(\mu^{\text{off}}), d_{\chi,o}(\mu^{\text{off}})\}.$$

Because rejection is weakly more valuable to the high type while the passing term is common, $O_h(\rho) \geq O_\ell(\rho)$. For $t \in \{0, p, 1\}$, write $a_t^{\text{pass}} = a_\chi^{\text{pass}}(t)$ and $d_{o,t} = d_{\chi,o}(t)$.

We can now enumerate the pure profiles. Under pooling, the posterior is p. If the common proposal passes, both types receive the same share z; feasibility requires $z \leq a_p^{\text{pass}}$, and absence of unsupported deviations requires $z \geq O_h(\rho)$. This gives the first row of Table 8. If the pooled proposal is rejected, type o receives $d_{o,p}$, so the two inequalities in the pooling-delay row are necessary and sufficient.

Under separation, Bayes' rule assigns posterior zero to the low-type proposal and posterior one to the high-type proposal. If both pass, either type can copy the other's proposal and receive the same share. Bilateral imitation therefore forces a common z. Feasibility and unsupported deviations give

$$O_h(\rho) \leq z \leq \min\{a_0^{\text{pass}}, a_1^{\text{pass}}\}.$$

If the low type passes and the high type delays, imitation, feasibility, and unsupported deviations require

$$\max\{d_{\ell,1}, O_\ell(\rho)\} \le z \le \min\{a_0^{\text{pass}}, d_{h,1}\}, \qquad d_{h,1} \ge O_h(\rho).$$

These are exactly the fourth row of the table. The reverse pattern is impossible: it would require $d_{\ell,0} \ge z \ge d_{h,0}$, although $d_{h,0} > d_{\ell,0}$. Finally, if both proposals are rejected, bilateral imitation gives $d_{\ell,0} \ge d_{\ell,1}$ and $d_{h,1} \ge d_{h,0}$; the remaining two inequalities in the last row exclude unsupported deviations. The pooling/separating and pass/reject enumeration is exhaustive. Canonical passing proposals and clearly rejected proposals attain every listed payoff, proving sufficiency as well as necessity.

For completeness with arbitrary Borel proposal laws $\sigma_\ell, \sigma_h \in \mathcal{P}(\mathcal{X})$, set

$$\bar\sigma = (1-p)\sigma_\ell + p\sigma_h, \qquad \mathsf{S}_M = \text{supp}(\bar\sigma).$$

Let $\hat\mu : \mathcal{X} \to [0,1]$ be the Borel posterior map and let $b_M : \mathcal{X} \to \{0,1\}$ be the scalar passage indicator generated by the full cutoff ballot vector. Define the Borel payoff maps

$$u_o(x) = b_M(x)x_H + \{1 - b_M(x)\}d_{\chi,o}(\hat\mu(x)), \qquad V_o = \int u_o(x)\,\sigma_o(dx).$$

Let u_o^{off} evaluate an unsupported proposal at μ^{off}, and put

$$\mathfrak{D}_o(\mathsf{S}_M) = \sup_{x \notin \mathsf{S}_M} u_o^{\text{off}}(x),$$

with the supremum of the empty set equal to $-\infty$. The proposal laws generate a PBE if and only if, in addition to Bayes, the ballot rule, and the continuation selection just specified,

$$\begin{aligned}
u_o(x) &\le V_o && \text{for every } x \in \mathsf{S}_M,\\
u_o(x) &= V_o && \text{for } \sigma_o\text{-almost every } x,\\
V_o &\ge \mathfrak{D}_o(\mathsf{S}_M), && o \in \{\ell, h\}.
\end{aligned}$$

Necessity follows because each type can copy every supported proposal and choose any unsupported proposal, while a mixed strategy assigns probability one to best responses. Conversely, prescribe the cutoff ballots and the continuation $\chi(\hat{\mu}(x))$. The first line rules out profitable supported deviations, the third rules out unsupported ones, and the second restricts each type's law to best responses. This proves sequential rationality at every proposal. The local Bayes ratio is a pointwise limit of Borel ratios on the closed support and is constant off it; hence $\hat{\mu}$, the finite cutoff ballot, and both payoff maps are Borel. The conditions are therefore also sufficient for the mixed correspondence stated in Appendix E.2.

At $p \in \{0, 1\}$, prior support fixes every posterior at the endpoint. Each type's complete Borel best-response correspondence is consequently the set of probability measures supported on its argmax between passage and rejection, with value $\overline{V}_o = \max\{a_p^{\text{pass}}, d_{o,p}\}$.

It remains to prove existence and the lower bound used in Appendix E.5. Let

$$v_M^{\text{safe}} = 1 - \frac{k\beta}{m}, \qquad T = \frac{v_M^{\text{safe}}}{\beta} = \frac{1}{\beta} - \frac{k}{m} > \frac{1}{m}.$$

If $h \leq T$, take $\rho = 1$ and pool on the canonical passing proposal at posterior p. To see that rejection is not profitable, consider the three possible continuation outcomes. Exclusion gives the high type at most $\beta h \leq \beta T = v_M^{\text{safe}}$; screening or pooling gives it at most $\beta^2 h < v_M^{\text{safe}}$, while the best passing proposal at that posterior leaves at least v_M^{safe}. The same inequalities hold for the low type and are preserved on the residual continuation segment by convexity.

If $\ell \leq T \leq h$, take $\mu^{\text{off}} = 1$. At posterior one, the majority continuation excludes H, so

$$d_{\ell,1} = \beta\ell \leq v_M^{\text{safe}} \leq \beta h = d_{h,1}.$$

At posterior zero, $a_0^{\text{pass}} \geq v_M^{\text{safe}}$. Choose $z = v_M^{\text{safe}}$: the low type passes with this share and the high type rejects. The displayed inequalities rule out imitation in both directions, and the posterior-one calculation also rules out unsupported deviations. If $T \leq \ell$, then $T > 1/m$ implies that the majority continuation excludes H at every posterior. Rejection gives each type $\beta o \geq v_M^{\text{safe}}$, so both types can pool on delay for any ρ. The adjacent constructions remain valid at equality; endpoints follow from the preceding argmax description. Thus a majority PBE exists for every admissible primitive and prior for some ρ.

Moreover, at any posterior H can pay β/m to any k weak states and retain v_M^{safe}. Since every cutoff is at most β/m, this proposal passes regardless of the belief it induces. A clearly rejected proposal gives type o at least $\beta^2 o$. Hence every majority equilibrium satisfies

$$V_M^A(o) \geq \max\{v_M^{\text{safe}}, \beta^2 o\},$$

which establishes the lower bound used by Proposition 6.2. □

B.8 Proof of the private-unanimity agenda correspondence

For an interior prior, let $\sigma_\ell, \sigma_h \in \mathcal{P}(\mathcal{X})$ be the two proposal laws, let $\bar{\sigma} = (1-p)\sigma_\ell + p\sigma_h$, and let $\mu : \mathcal{X} \to [0,1]$ be the public posterior map. Write

$$r_U(o) = \frac{\beta(1-\beta o)}{m}, \qquad z_L = v_U^A(\ell) = 1 - \beta + \beta^2 \ell,$$
$$z_H = v_U^A(h) = 1 - \beta + \beta^2 h, \quad d = \beta^2 h, \qquad \Delta = z_L - d.$$

The associated cutoff proposals are

$$x^\ell = (z_L, r_U(\ell), \dots, r_U(\ell)), \qquad x^h = (z_H, r_U(h), \dots, r_U(h)).$$

At posterior zero, unanimity prices every weak vote at $r_U(\ell)$; at a posterior above p^*, it prices every weak vote at $r_U(h)$. Since one no vote rejects the proposal, sequential rationality after that vote requires a continuation at the same posterior. The baseline unanimity correspondence is empty on $(0, p^*]$. Therefore every disciplined posterior and the common off-support posterior must lie in $\mathcal{P}_C = \{0\} \cup (p^*, 1]$.

The cutoff sets are closed because equality is accepted. The maximal shares that H can retain under the two admissible prices are attained at x^ℓ and x^h, and equal z_L and z_H, respectively. Rejection gives the high type d at every admissible posterior. It gives the low type $\beta^2\ell$ at posterior zero and d at a high posterior. In particular,

$$0 < r_U(h) < r_U(\ell), \qquad z_H = z_L + \beta^2(h - \ell) > z_L, \qquad z_H > d.$$

We first establish two reduction lemmas. For an interior prior, the two types receive the same equilibrium payoff. For σ_h-almost every proposal, the posterior is high. If it passes, the low type can copy it and obtain the same share; if it rejects, both types receive d. A positive-mass zero-posterior signal can be used only by the low type. It cannot reject in equilibrium: the high type can guarantee d, and the low type can copy a best high-type signal, whereas rejection at zero gives it only $\beta^2\ell < d$. It must therefore pass, after which the high type can copy it and obtain the same share. These bilateral imitation inequalities, together with equality to the type's value σ_o-almost everywhere, imply a common payoff V. Pointwise deviation inequalities continue to govern disciplined support points that carry zero mass.

Next let

$$\lambda_0 = \int 1\{\mu(x) = 0\}\, d\bar{\sigma}(x)$$

be the public mass of zero-posterior proposals. If $\lambda_0 > 0$, the low type uses such a proposal and can receive at most z_L; only x^ℓ attains that maximum. The common-payoff result and the high type's rejection option require $z_L \geq d$, or $\Delta \geq 0$. An off-support high posterior would permit the deviation x^h, which yields $z_H > z_L$; hence $\mu^{\text{off}} = 0$. It follows that $V = z_L$, $\sigma_\ell(\{x^\ell\}) > 0$, and $\sigma_h(\{x^\ell\}) = 0$. If instead $\lambda_0 = 0$, Bayes plausibility gives

$$p = \int \mu(x)\, d\bar{\sigma}(x) > p^*,$$

because the posterior is above p^* almost surely. These two cases are exhaustive.

Suppose first that $\lambda_0 > 0$. The preceding argument gives the domain of the low-posterior family. For σ_o-almost every high-posterior proposal used by type o, the proposal must either pass with H's share z_L, or, only when $z_L = d$, reject and deliver the same payoff. The pointwise deviation restrictions in Appendix E.3 are necessary because either type can choose every proposal, including supported limit points of zero mass. They are sufficient because all used proposals then yield z_L, while every other disciplined or unsupported proposal yields at most z_L. A pure witness is

$$\sigma_\ell = \delta_{x^\ell}, \qquad \sigma_h = \delta_{x^S}, \qquad x^S = (z_L, r_U(h), \dots, r_U(h)).$$

The proposal x^S passes and is feasible because $z_L + m r_U(h) = 1 - \beta^2(h - \ell) < 1$. Neither type gains by imitation or by rejection precisely when $\Delta \geq 0$. This proves necessity and sufficiency of the low-posterior family.

Now suppose $\lambda_0 = 0$, so $p > p^*$. If $\mu^{\text{off}} = 0$, the common payoff must satisfy

$$\max\{z_L, d\} \leq V \leq z_H.$$

The lower bound excludes the best zero-posterior passing and rejection deviations; the upper bound is the maximal feasible share at a high posterior. For every V in the interval, the pooling proposal

$$x(V) = \left(V, \frac{1-V}{m}, \dots, \frac{1-V}{m}\right)$$

passes because $V \leq z_H$ implies $(1-V)/m \geq r_U(h)$. It gives posterior p and leaves no profitable off-support deviation because $V \geq \max\{z_L, d\}$. This proves existence of every payoff in the high-posterior family with zero off-support belief; the pointwise restrictions in Appendix E.3 give the corresponding necessary and sufficient condition for arbitrary Borel laws.

If $\mu^{\text{off}} > p^*$, the off-support proposal x^h guarantees z_H. Feasibility also bounds the common payoff by z_H; hence $V = z_H$. The only passing proposal that leaves this share is x^h, so both proposal laws are δ_{x^h}. An off-support posterior in $(0, p^*]$ is inadmissible because it has no baseline continuation. Thus the two high-posterior subfamilies in Appendix E.3 are exhaustive. Combining them with the dichotomy for λ_0 proves that there is no third interior family and gives all stated existence cells.

At $p = 0$, support preservation fixes every posterior at zero. The low type uniquely chooses x^ℓ. The counterfactual high type chooses x^ℓ if $z_L > d$, may mix x^ℓ with rejected proposals if $z_L = d$, and chooses any rejected best response if $z_L < d$. This gives the endpoint correspondence in Appendix E.3. At $p = 1$, every posterior is one and both types uniquely choose x^h, since $z_H = d+1-\beta > d$.

Finally, any passing unanimity proposal gives every weak state at least $r_U(h)$, so it leaves H at most z_H; every rejection payoff is at most $d < z_H$. Therefore every unanimity assessment satisfies

$$V_U^A(o) \leq z_H = v_U^A(h).$$

Together with the majority lower bound in Appendix B.7, this proves the two bounds used in Appendix E.5. The linked payoff image, informational-rent incidence, and total agenda-effect cells then follow by the fixed-vector subtractions displayed in Appendices E.9 and E.13. □

B.9 Proof of the exact and economic representations

The argument is the same for majority and unanimity, so fix $g \in \{M, U\}$ and a complete equilibrium assessment R_g. Let Z_g be the rule-specific realized-outcome space whose coordinates are the named proposal, realized posterior, passage indicator, continuation state, and terminal outcome tuple. It is a finite product and disjoint union of compact Polish spaces and hence is compact Polish; Appendices E.2 and E.3 give the concrete constructions. Let $\Gamma_o^{g,R_g} \in \mathcal{P}(Z_g)$ be the resulting type-o joint law, and write $\Gamma_o = \Gamma_o^{g,R_g}$ within this proof. A common permutation $\tau \in G = S_m$ of weak-state names acts on Z_g by a homeomorphism T_τ. Relabeling a PBE by T_τ preserves feasibility and payoffs, transports relative balls and supports, commutes with the local Bayes limit, and maps ballots, continuation kernels, and deviations bijectively to payoff-equivalent objects. The relabeled object is therefore also a PBE.

Set $X_g = \mathcal{P}(Z_g)^2$, with the weak topology, and let the finite group act diagonally on the pair of type-specific realized laws. This is a Polish space with a continuous group action. For $x \in X_g$, define the orbit law

$$\Lambda_x = \frac{1}{|G|} \sum_{\tau \in G} \delta_{\tau x}.$$

For every Borel set $C \subseteq X_g$,

$$\Lambda_x(C) = \frac{1}{|G|} \sum_{\tau \in G} 1_C(\tau x)$$

is Borel in x, so $x \mapsto \Lambda_x$ is Borel. Right multiplication only reorders the finite sum, proving invariance. If $x' = \tau x$, the two orbit laws coincide. Conversely, if $\Lambda_x = \Lambda_{x'}$, the Borel singleton $\{x'\}$ has positive mass under $\Lambda_{x'}$, hence also under Λ_x; therefore $x' = \tau x$ for some τ. The orbit law is thus a complete invariant of diagonal relabeling, which proves the claim about Sig_g^{ex}.

For the economic layer, choose a Borel injection of Z_g into $[0, 1]$ and let q_g map each realized tuple to the least element of its finite orbit. This gives a Borel transversal and a Borel quotient map satisfying $q_g \circ T_\tau = q_g$. If E is a standard Borel space and $f : Z_g \to E$ is Borel and invariant, define $\bar{f}$ on the transversal by its value at the selected representative. Invariance makes the definition independent of the original named tuple, and surjectivity of q_g gives uniqueness. Thus

$$f = \bar{f} \circ q_g.$$

If, in addition, f is real-valued and integrable, then

$$\int f \, d\Gamma_o = \int \bar{f} \, d(q_g)_{\#}\Gamma_o.$$

Payoffs, passage and delay, the posterior law, anonymous terminal outcomes, and the ordered multiset of weak-state payoffs are invariant Borel maps, so they factor through the economic summary. The same is true of the payoff and outcome comparisons after the complete majority and unanimity assessments have first been paired at the same primitives and off-path belief specification. Named proposals, named supports, and off-path strategy or belief functions remain in the underlying complete assessment. This proves the measurability, orbit-completeness, and universal anonymous factorization statements in Appendix F.1. □

Appendix C: Endpoints, exact sets, and illustration

C.1 Endpoint equivalence

At $p = 0$, posterior support is the singleton low type. The private unanimity proposal and payoff coincide with the public low-type game. At $p = 1$, posterior support is the singleton high type, and the private pooling proposal and payoff coincide with the public high-type game. The same equivalence holds under majority: applying the relevant private class at each endpoint reproduces public inclusion or exclusion, including the $o = 1/m$ tie-break. Thus the endpoints are literal complete-information games, not one-sided limits.

C.2 Exact correspondence and envelopes

Most parameter cells are singleton payoff vectors. At $h = 1/m$, exact indifference between exclusion and pooling can leave a proposal segment. If λ is the exclusion weight, every reported object uses that same λ. The lower and upper envelope of any component may summarize the segment, but coordinated marginal envelopes do not imply that all pairwise combinations are attainable.

For majority-rule informational rents, the exact componentwise envelopes are

$$IR_M^B(\ell) \in [\min\{(1-\beta)\ell, \beta(h-\ell)\}, \max\{(1-\beta)\ell, \beta(h-\ell)\}], \qquad IR_M^B(h) \in [0, (1-\beta)h].$$

For both the private institutional payoff contrast and ΔIR^B, they are

$$\text{low type} \in [\min\{0, \beta h - \ell\}, \max\{0, \beta h - \ell\}], \qquad \text{high type} \in [-(1-\beta)h, 0].$$

In each case these are the marginal envelopes of a single line segment. The same λ indexes both coordinates, so the Cartesian product of the two intervals is not attainable.

Where $0 < p \leq p^*$, the private unanimity correspondence is empty. Accordingly, V_U^B, IR_U^B, and ΔIR^B are empty. No payoff, sign, institutional ranking, or interpolation is assigned to that cell.

Appendix D: Notation

Table 7: Notation for the bargaining game

Symbol	Meaning
H	Privately informed hegemon
W, m	Set and number of weak states; $m \geq 3$
$k = \lfloor (m+1)/2 \rfloor$	Additional yes votes a proposer needs under majority
$o \in \{\ell, h\}$	Hegemon's terminal disagreement payoff, with $0 < \ell < h < 1$
p	Prior probability of the high disagreement payoff h
μ, μ^{off}	Posterior probability of h, on and off path
β	One-round discount factor, strictly between zero and one
$x = (x_H, x_1, \ldots, x_m)$	Feasible allocation vector
x_H, x_j, x_i	Allocations to H, weak responder j, and proposer i
$w = \beta/m$	First-round continuation value of a weak state under majority
$w_o^U = \beta(1-o)/m$	First-round continuation value of a weak state under unanimity
p^*	Terminal unanimity cutoff, $(h-\ell)/(1-\ell)$
B, A	Games without and with the earlier hegemonic agenda stage
$v_g^B(o)$, $v_g^A(o)$	Public-information payoff under rule g in each arm
$V_g^B(o)$, $V_g^A(o)$	Private-information payoff correspondence in each arm
IR_g^B, IR_g^A	Private payoff minus the same-arm public benchmark
ΔIR^B, ΔIR^A	Unanimity-minus-majority informational-rent contrasts
$e = m - k$	Weak states excluded by a minimal majority coalition
v_M^{safe}	Safe majority-agenda payoff, $1 - k\beta/m$
o_M^*	Majority-agenda delay cutoff, v_M^{safe}/β
ρ	Off-path likelihood-ratio coordinate in the agenda extension
$\hat{\mu}(\cdot)$	Borel posterior map in mixed majority assessments
R_g	Complete equilibrium assessment under rule g, including strategies, beliefs, continuation selections, and induced outcome laws

Symbol	Meaning
$\mathcal{B}_g(\mathbf{d}, \rho, \mu^{\text{off}})$	Correspondence of complete assessments at a fixed primitive and off-path belief specification
$\mathcal{J}_A^{\text{cmp}}$	Restricted product pairing majority and unanimity assessments at the same specification
Γ_o^{g,R_g}	Type-specific joint realized law induced by assessment R_g
Sig_g^{ex}, Sum_g^{econ}	Exact relabeling signature and anonymous economic summary
ea	Ex ante coordinate, evaluated as the prior-weighted average of the two types
$\Delta v^A(o)$	Public institutional gap with agenda, $v_U^A(o) - v_M^A(o)$
$\Delta V^A(o)$	Private institutional contrast with agenda, $V_U^A(o) - V_M^A(o)$
D_g, I_g, T_g	Public agenda difference, agenda–information component, and total private agenda difference
Q_g	Composite diagonal comparison

The symbols w and w_o^U price a weak vote inside the baseline arm. The functions $r_M(o)$ and $r_U(o)$ in Appendix E price a weak vote one date earlier, after rejection enters the baseline continuation. The extra discounting means that these are date-specific objects, not alternative names for the same quantity.

Appendix E: Exact agenda-stage correspondences and comparisons

E.1 Agenda-stage contract

Fix

$$\mathbf{d} = (m, k, e, \beta, \ell, h, \mathcal{X}, p),$$

$$m \geq 3, \quad k = \lfloor (m+1)/2 \rfloor, \quad e = m - k,$$

$$0 < \beta < 1, \quad 0 < \ell < h < 1, \quad p \in [0, 1].$$

At date A, after observing its type, H proposes

$$x^A = (x_H, x_1, \dots, x_m) \in \mathcal{X}, \qquad \mathcal{X} = \{x^A : x_H \geq 0,\ x_j \geq 0,\ x_H + \textstyle\sum_j x_j \leq 1\}.$$

The proposal counts as H's affirmative vote. Weak states vote simultaneously and in pure strategies. Majority requires k affirmative weak votes; unanimity requires all m. Passage implements the package at date A. Rejection enters the corresponding specified Round-1 continuation, whose native payoff is multiplied by β exactly once.

A rejected public history includes the rule, proposal, proposer, full ballot, outcome, and posterior. Its continuation selector is total, public, Borel, common to types compatible with the history, and returns one complete Round-1 equilibrium assessment. That assessment jointly specifies strategies, beliefs, recognition lotteries, ballots, payoffs, and outcomes; the continuation object is therefore the full assessment rather than a scalar payoff.

For an interior prior, the equilibrium proposal laws of the two types are Borel measures σ_ℓ, σ_h. At every disciplined proposal, Bayes' rule determines the posterior from their local likelihood ratio. At undisciplined proposals, the common off-path belief restriction fixes the posterior to

$$\mu^{\text{off}} = b_\rho(p) = \frac{p\rho}{1 - p + p\rho}, \qquad \rho \in [0, \infty].$$

The extended-real boundary conventions are

$$b_0(p) = 0, \qquad b_\infty(p) = 1$$

for an interior prior. The endpoints preserve prior support: $\mu^{\text{off}} = p$ everywhere, the comparison index is $(*, p)$, and ρ is irrelevant. Voting follows the same as-if-pivotal comparison and indifference-to-yes convention as the baseline.

The single coordinate ρ is common to every undisciplined proposal. This is a stronger restriction than the ballot-by-ballot free values of Appendix A.2, adopted here to index the correspondences. The two conventions apply at different levels: a rejected agenda proposal enters a complete Round-1 assessment whose internal beliefs follow the baseline rule.

E.2 Complete private-majority correspondence

For each posterior μ and a fixed anonymous Borel Markov continuation selection $\chi : [0,1] \to \mathfrak{C}_M$, let $a_\chi^{\text{pass}}(\mu)$ be H's best payoff from a proposal that passes and let $d_{\chi,o}(\mu)$ be the payoff of a type with disagreement payoff o from rejection, both at date A. Define

$$a_\mu^{\text{pass}} := a_\chi^{\text{pass}}(\mu), \qquad d_{o,\mu} := d_{\chi,o}(\mu), \qquad \mu \in [0,1],$$

so that $a_p^{\text{pass}}, a_0^{\text{pass}}, a_1^{\text{pass}}$ and $d_{\ell,p}, d_{h,p}, d_{\ell,0}, d_{\ell,1}, d_{h,0}, d_{h,1}$ denote evaluations of these same functions at the displayed posterior. Also define

$$O_o(\rho) = \max\{a_\chi^{\text{pass}}(b_\rho(p)), d_{\chi,o}(b_\rho(p))\}.$$

The following conditions are necessary and sufficient for every interior pure equilibrium form:

Table 8: Pure private-majority equilibrium forms at the agenda stage

Form	Necessary and sufficient condition and linked type payoff
Pooling agreement	$O_h(\rho) \le a_p^{\text{pass}}$, with any $z \in [O_h(\rho), a_p^{\text{pass}}]$; linked payoff (z,z)
Pooling delay	$d_{\ell,p} \ge O_\ell(\rho)$ and $d_{h,p} \ge O_h(\rho)$; linked payoff $(d_{\ell,p}, d_{h,p})$
Separating agreement–agreement	$O_h(\rho) \le \min\{a_0^{\text{pass}}, a_1^{\text{pass}}\}$, with any $z \in [O_h(\rho), \min\{a_0^{\text{pass}}, a_1^{\text{pass}}\}]$; linked payoff (z,z)
Low agreement, high delay	$d_{h,1} \ge O_h(\rho)$ and $\max\{d_{\ell,1}, O_\ell(\rho)\} \le \min\{a_0^{\text{pass}}, d_{h,1}\}$; linked payoff $$(z, d_{h,1}), \qquad z \in [\max\{d_{\ell,1}, O_\ell(\rho)\}, \min\{a_0^{\text{pass}}, d_{h,1}\}]$$
Low delay, high agreement	Impossible; linked payoff $\emptyset$
Separating delay–delay	$d_{\ell,0} \ge d_{\ell,1}$, $d_{h,1} \ge d_{h,0}$, $d_{\ell,0} \ge O_\ell(\rho)$, and $d_{h,1} \ge O_h(\rho)$; linked payoff $(d_{\ell,0}, d_{h,1})$

These forms preserve proposals and messages even when their payoffs coincide. Separating agreement at the same z is separation in messages at indifference, not pooling. The off-path feasibility set for ρ can be disconnected because the majority continuation changes branches. The table therefore supplies the relevant branch-specific conditions.

For mixed strategies, a complete equilibrium assessment is

$$R_M = (\rho, \mu^{\text{off}}, \sigma_\ell, \sigma_h, \bar{\sigma}, \hat{\mu}, \chi, b_M, u_\ell, u_h).$$

Here $\sigma_\ell, \sigma_h, \bar{\sigma} \in \mathcal{P}(\mathcal{X})$, with $\bar{\sigma} = (1-p)\sigma_\ell + p\sigma_h$, and $\hat{\mu} : \mathcal{X} \to [0,1]$ is the Borel posterior map. The scalar map $b_M : \mathcal{X} \to \{0,1\}$ is the passage indicator generated by the full cutoff ballot vector. The assessment also retains the continuation selection χ and the Borel payoff maps $u_\ell, u_h : \mathcal{X} \to \mathbb{R}$. Let

$$\begin{aligned} \mathsf{S}_M = \operatorname{supp}(\bar{\sigma}), \qquad & u_o(x) = b_M(x) x_H + \{1 - b_M(x)\} d_{\chi,o}(\hat{\mu}(x)), \\ V_o = \int u_o(x)\, \sigma_o(dx), \qquad & \mathfrak{O}_o(\mathsf{S}_M) = \sup_{x \notin \mathsf{S}_M} u_o^{\text{off}}(x), \end{aligned}$$

where u_o^{off} evaluates an unsupported proposal at μ^{off}, and the supremum of the empty set is $-\infty$. In addition to pointwise Bayes updating on the disciplined support, the pivotal ballot rule at every proposal, and the continuation selection χ, the necessary-and-sufficient proposal conditions are

$$\begin{aligned} u_o(x) \le V_o \qquad & \text{for every } x \in \mathsf{S}_M, \\ u_o(x) = V_o \qquad & \text{for } \sigma_o\text{-almost every } x, \\ V_o \ge \mathfrak{O}_o(\mathsf{S}_M), \quad & o \in \{\ell, h\}. \end{aligned}$$

Thus arbitrary Borel mixtures and atomless supports are included.

At $p \in \{0,1\}$, the posterior is fixed at the endpoint. With

$$\overline{V}_o = \max\{a_p^{\text{pass}}, d_{o,p}\},$$

each type's Borel proposal law may be any probability measure supported on its argmax. The linked type-payoff vector retains the probability-zero type. For every admissible primitive and prior, at least one majority PBE exists for some ρ; this does not assert existence at every fixed ρ. Appendix B.7 proves the pure enumeration, the mixed membership criterion, the endpoint correspondences, existence, and the uniform lower bound used in the body.

The realized-law layer is constructed as follows. For fixed primitives, let Ω_D^M be the finite discrete set of literal terminal continuation tuples generated by the admissible uniform majority kernels: each tuple retains the recognized proposer, coalition, proposal, votes, and terminal outcome. Define the compact Polish spaces

$$\Omega_T^M = (\{\mathsf{A}\} \times \mathcal{X}) \sqcup (\{\mathsf{D}\} \times \Omega_D^M), \qquad Z_M = \mathcal{X} \times [0,1] \times \{0,1\} \times \mathfrak{C}_M \times \Omega_T^M.$$

For $\xi \in \mathfrak{C}_M$, let $K_{o,\xi}^D \in \mathcal{P}(\Omega_D^M)$ be the selected literal continuation kernel and let $\widetilde{K}_{o,\xi}^D$ be its pushforward to $\{\mathsf{D}\} \times \Omega_D^M$. Conditional on proposal x, set

$$L_o^{R_M}(d\omega_T \mid x) = b_M(x)\delta_{(\mathsf{A},x)}(d\omega_T) + \{1 - b_M(x)\}\widetilde{K}_{o,\chi(\widehat{\mu}(x))}^D(d\omega_T).$$

The type-specific joint realized law is therefore

$$\Gamma_o^{M,R_M}(C) = \int_{\mathcal{X}} \int_{\Omega_T^M} 1_C(x, \hat{\mu}(x), b_M(x), \chi(\hat{\mu}(x)), \omega_T)\, L_o^{R_M}(d\omega_T \mid x)\, \sigma_o(dx).$$

Thus $\Gamma_o^{M,R_M} \in \mathcal{P}(Z_M)$. This single law jointly determines the named proposal, posterior, passage, literal continuation selection, and terminal outcome.

Let $G = S_m$ act on Z_M by applying the same permutation to all named weak-state coordinates while fixing the posterior, passage indicator, and continuation label. Let T_τ^M denote this action and let $q_M : Z_M \to Z_M/G$ send each realized tuple to the selected representative of its finite orbit. For $\gamma_M = (\Gamma_\ell^{M,R_M}, \Gamma_h^{M,R_M})$, let Λ_{γ_M} be the uniform law of its diagonal G-orbit. Thus, for an interior prior,

$$\mathrm{Sig}_M^{ex}(R_M) = (\rho, \mu^{\mathrm{off}}, \Lambda_{\gamma_M}), \qquad \mathrm{Sum}_M^{econ}(R_M) = (\rho, \mu^{\mathrm{off}}, (q_M)_\# \Gamma_\ell^{M,R_M}, (q_M)_\# \Gamma_h^{M,R_M}),$$

with endpoint forms obtained by replacing $(\rho, \mu^{\mathrm{off}})$ with $(*, p)$. The first object preserves the complete diagonal orbit of the two realized laws; the second removes weak names tuple by tuple.

E.3 Complete private-unanimity correspondence

Define

$$r_U(o) = \frac{\beta(1-\beta o)}{m}, \qquad v_U^A(o) = 1 - \beta + \beta^2 o.$$

Thus $0 < r_U(h) < r_U(\ell)$, and the two proposal primitives are

$$x^\ell = (v_U^A(\ell), r_U(\ell), \dots, r_U(\ell)), \qquad x^h = (v_U^A(h), r_U(h), \dots, r_U(h)).$$

At posterior zero, every weak state accepts exactly when $x_j \geq r_U(\ell)$; at a high posterior, it accepts exactly when $x_j \geq r_U(h)$. Equality passes under the indifference-to-yes convention. The only continuation posteriors admitted by the specified unanimity continuation are

$$\mathcal{P}_C = \{0\} \cup (p^*, 1].$$

Rejection pays the high type $\beta^2 h$ at every admissible posterior; it pays the low type $\beta^2 \ell$ at posterior zero and $\beta^2 h$ at a high posterior.

A complete unanimity equilibrium assessment is

$$R_U = (\sigma_\ell, \sigma_h, \mu, \mu^{\text{off}}, \hat{\kappa}_U, b_U, \Omega, \Gamma_\ell, \Gamma_h).$$

Here σ_o is a Borel proposal law, $\bar{\sigma} = (1-p)\sigma_\ell + p\sigma_h$, μ is the public posterior, and $\mu^{\text{off}} \in \mathcal{P}_C$ is its single value at every undisciplined proposal. The selector $\hat{\kappa}_U$ returns a complete literal continuation assessment at the same posterior, $b_U : \mathcal{X} \to \{\mathsf{Y}, \mathsf{N}\}^m$ is the full pure ballot map, and $\Omega_o(x)$ is the literal terminal law induced by passage or by that continuation. Finally, Γ_o is the five-coordinate realized pushforward law constructed below. The ballot map, literal selector, and terminal laws are jointly generated by the same assessment. The local Bayes limit must exist at every disciplined proposal, including zero-mass limit points, and must lie in $\mathcal{P}_C$. At each such point, and at every undisciplined point evaluated with μ^{off}, neither type may obtain a profitable proposal deviation. These pointwise restrictions, the ballot map, the literal continuation, and both realized laws are components of that same assessment.

The full interior correspondence is generated by the following two families.

Low-posterior family. Its exact domain is

$$0 < p < 1, \qquad v_U^A(\ell) \geq \beta^2 h, \qquad \mu^{\text{off}} = 0.$$

Choose Borel laws satisfying all of the following:

1. $\sigma_\ell(\{x^\ell\}) = \alpha_L > 0$ and $\sigma_h(\{x^\ell\}) = 0$;
2. outside x^ℓ, $\mu(x) > p^*$ $\bar{\sigma}$-almost surely;
3. each used proposal outside x^ℓ either passes with H's share $v_U^A(\ell)$, or, only when $v_U^A(\ell) = \beta^2 h$, rejects and pays $\beta^2 h = v_U^A(\ell)$;
4. every disciplined proposal, including every zero-mass limit point, gives each type a deviation payoff no larger than $v_U^A(\ell)$; every undisciplined proposal satisfies the same inequality under posterior zero.

Every accepted high-posterior signal therefore belongs to

$$\mathcal{K}_L = \{x \in \mathcal{X} : x_H = v_U^A(\ell),\ \min_j x_j \geq r_U(h)\}.$$

The laws may be asymmetric, discrete, semi-pooling, mixed, or atomless. A pure witness has the low type propose x^ℓ and the high type propose $x^S = (v_U^A(\ell), r_U(h), \dots, r_U(h))$. Every assessment has linked payoff $(v_U^A(\ell), v_U^A(\ell))$. If $v_U^A(\ell) > \beta^2 h$, agreement occurs with probability one for both types; if $v_U^A(\ell) = \beta^2 h$, high-posterior rejected signals may carry mass. The weak-state payoff attached to the same assessment is x_j after passage, $r_U(\ell)$ after rejection at $(\mu, o) = (0, \ell)$, zero after rejection at $(0, h)$, and $r_U(h)$ after rejection at any high posterior.

High-posterior family. Its necessary prior domain is $p^* < p < 1$. When $\mu^{\text{off}} = 0$, choose any

$$u \in [\max\{v_U^A(\ell), \beta^2 h\}, v_U^A(h)],$$

and Borel laws such that: $\mu > p^*$ $\bar{\sigma}$-almost surely; the local admissible Bayes limit exists at every disciplined point; every used accepted proposal lies in

$$\mathcal{K}_H(u) = \{x \in \mathcal{X} : x_H = u,\ \min_j x_j \geq r_U(h)\};$$

used rejected proposals occur only when $u = \beta^2 h$; every disciplined deviation gives each type at most u; and every undisciplined deviation, evaluated at posterior zero, also gives at most u. The linked payoff is (u, u). If $u > \beta^2 h$, agreement has probability one for each type; if $u = \beta^2 h$, each type may mix accepted proposals with $x_H = \beta^2 h$ and rejected high-posterior proposals. A pure pooling witness for every u is

$$x(u) = \left(u, \frac{1-u}{m}, \dots, \frac{1-u}{m}\right), \qquad \sigma_\ell = \sigma_h = \delta_{x(u)}, \qquad \mu(x(u)) = p.$$

When $\mu^{\text{off}} \in (p^*, 1]$, the only current proposal laws are $\sigma_\ell = \sigma_h = \delta_{x^h}$, with $\mu(x^h) = p$ and payoff $(v_U^A(h), v_U^A(h))$; multiplicity survives only in μ^{off} and the complete literal continuation assessment. When $\mu^{\text{off}} \in (0, p^*]$, the correspondence is empty. For $p^* < p < 1$ and $\mu^{\text{off}} = 0$, the high-posterior family always exists, and the low-posterior family is additionally present exactly when $v_U^A(\ell) \geq \beta^2 h$. For $0 < p \leq p^*$, equilibrium exists exactly when $v_U^A(\ell) \geq \beta^2 h$ and consists only of the low-posterior family. There is no third interior family.

The endpoints are complete Borel strategy correspondences, not one-sided limits. At $p = 0$, support preservation fixes $\mu(x) = \mu^{\text{off}} = 0$ everywhere, $\sigma_\ell = \delta_{x^\ell}$, and the counterfactual high-type law is

$$\begin{cases} \delta_{x^\ell}, & v_U^A(\ell) > \beta^2 h, \\ \text{any Borel law supported on } \{x^\ell\} \cup \mathcal{R}_L, & v_U^A(\ell) = \beta^2 h, \\ \text{any Borel law supported on } \mathcal{R}_L, & v_U^A(\ell) < \beta^2 h, \end{cases}$$

where

$$\mathcal{R}_L = \{x \in \mathcal{X} : \min_j x_j < r_U(\ell)\}.$$

The linked payoff is $(v_U^A(\ell), \max\{v_U^A(\ell), \beta^2 h\})$, and the low type agrees immediately with probability one. At $p = 1$, support preservation fixes $\mu(x) = \mu^{\text{off}} = 1$ everywhere and $\sigma_\ell = \sigma_h = \delta_{x^h}$, including the probability-zero type's strategy, with linked payoff

$$V_U^A = (v_U^A(h), v_U^A(h)).$$

Consequently, the payoff image of the complete correspondence is

$$V_U^A = \begin{cases} (v_U^A(\ell), \max\{v_U^A(\ell), \beta^2 h\}), & p = 0, \\ (v_U^A(\ell), v_U^A(\ell)), & 0 < p \le p^*,\ v_U^A(\ell) \ge \beta^2 h,\ \mu^{\text{off}} = 0, \\ \{(u,u) : u \in [\max\{v_U^A(\ell), \beta^2 h\}, v_U^A(h)]\}, & p^* < p < 1,\ \mu^{\text{off}} = 0, \\ (v_U^A(h), v_U^A(h)), & p^* < p < 1,\ \mu^{\text{off}} \in (p^*, 1], \\ (v_U^A(h), v_U^A(h)), & p = 1, \\ \emptyset, & \text{all remaining parameter and belief specifications.} \end{cases}$$

This payoff image preserves each assessment's linked vector; an interval of such vectors remains a one-dimensional set.

For the two-layer representation, let $\mathfrak{C}_U = \{\mathsf{L}, \mathsf{P}\}$ be the discrete space of literal unanimity-continuation cells. Let Ω_D^U be the compact space of terminal continuation tuples generated by those cells: each tuple retains the recognized weak state, allocation, vote vector, outcome, and payoff vector. Define

$$\Omega_T^U = (\{\mathsf{A}\} \times \mathcal{X}) \sqcup (\{\mathsf{D}\} \times \Omega_D^U), \qquad Z_U = \mathcal{X} \times [0,1] \times \{0,1\} \times \mathfrak{C}_U \times \Omega_T^U.$$

These are compact Polish spaces. Set

$$a_U(x) = 1\{b_U(x) = (\mathsf{Y}, \dots, \mathsf{Y})\}, \qquad \xi_U(x) = \text{cell}(\hat{\kappa}_U(\mu(x))) \in \mathfrak{C}_U,$$

and let ω_T be the terminal outcome tuple induced by $\Omega_o(x)$. Then

$$\Gamma_o^{U,R_U} = \text{Law}_o(x, \mu(x), a_U(x), \xi_U(x), \omega_T) \in \mathcal{P}(Z_U).$$

Let $q_U : Z_U \to Z_U/G$ send every realized tuple to the selected representative of its finite weak-name orbit. Under a common permutation τ of weak-state names, let $\mathcal{R}_\tau$ relabel every named weak-state coordinate and let Λ_γ code the diagonal orbit of $\gamma = (\Gamma_\ell, \Gamma_h)$. For interior priors,

$$\mathrm{Sig}_U^{ex}(R_U) = (\rho, \mu^{\mathrm{off}}, \Lambda_\gamma), \qquad \mathrm{Sum}_U^{econ}(R_U) = (\rho, \mu^{\mathrm{off}}, (q_U)_\# \Gamma_\ell, (q_U)_\# \Gamma_h),$$

with endpoint forms $(*, p, \Lambda_\gamma)$ and $(*, p, (q_U)_\# \Gamma_\ell, (q_U)_\# \Gamma_h)$. The exact signature preserves the diagonal orbit of realized laws and their relevant realized identities; the economic layer removes names tuple by tuple. The full off-path functions reside in the complete equilibrium assessment. Formal-identity and realized-law operations use the exact signature; operations that depend on off-path beliefs, strategies, supports, messages, or continuation plans use the complete assessment. The orbit law serves only as an invariant index of relabeling equivalence, and empty parameter cells map to empty sets in both layers. Appendix B.8 proves the characterization, exhaustiveness, existence domains, endpoint correspondences, and upper payoff bound.

E.4 Exact institutional comparison at a fixed specification

Let $\mathcal{B}_M(\mathbf{d}, \rho, \mu^{\mathrm{off}})$ and $\mathcal{B}_U(\mathbf{d}, \rho, \mu^{\mathrm{off}})$ be the full correspondences just described. The admissible comparison set is

$$\mathcal{J}_A^{\mathrm{cmp}}(\mathbf{d}, \rho, \mu^{\mathrm{off}}) = \mathcal{B}_M(\mathbf{d}, \rho, \mu^{\mathrm{off}}) \times_{(\mathbf{d}, \rho, \mu^{\mathrm{off}})} \mathcal{B}_U(\mathbf{d}, \rho, \mu^{\mathrm{off}}).$$

This product pairs assessments with identical primitives and the same off-path belief specification. Proposals and continuation realizations remain rule-specific. The exact linked type-payoff contrast is

$$\mathcal{C}_{\ell h}^A(\mathbf{d}, \rho, \mu^{\mathrm{off}}) = \{u - v : u \in V_U^A(\mathbf{d}, \rho, \mu^{\mathrm{off}}),\ v \in V_M^A(\mathbf{d}, \rho, \mu^{\mathrm{off}})\}.$$

It is a difference of complete vectors. The ex ante image is

$$\mathcal{C}_{\mathrm{ea}}^A(\mathbf{d}, \rho, \mu^{\mathrm{off}}) = \{(1-p)x_\ell + p x_h : (x_\ell, x_h) \in \mathcal{C}_{\ell h}^A(\mathbf{d}, \rho, \mu^{\mathrm{off}})\}.$$

The ex ante image applies one prior to each complete linked vector. Dominance claims are setwise: unanimity dominates in a coordinate only when every assessment has positive sign; majority dominates only when every assessment has negative sign. A crossing set is selection dependent. An empty comparison set has no institutional ranking.

E.5 The sufficient majority-advantage region

Every majority assessment satisfies

$$V_M^A(o) \geq \max\{v_M^{\text{safe}}, \beta^2 o\}, \qquad v_M^{\text{safe}} = 1 - \frac{k\beta}{m},$$

while every unanimity assessment satisfies

$$V_U^A(o) \leq v_U^A(h) = 1 - \beta + \beta^2 h.$$

Therefore

$$v_M^{\text{safe}} - v_U^A(h) = \beta\left(\frac{e}{m} - \beta h\right).$$

If $\beta h < e/m$, then this margin is positive and every comparable assessment satisfies

$$V_U^A(o) - V_M^A(o) \leq -\beta\left(\frac{e}{m} - \beta h\right) < 0$$

for both types and ex ante. Equality yields weak majority advantage. This proves sufficiency only.

The condition is not necessary. The following separate parameter point is chosen to demonstrate that failure of necessity, rather than to replace the running illustration. With

$$m = 4,\ k = 2,\ e = 2,\ \beta = .9, \ell = .5,\ h = .6,\ p = 0,$$

we have $e/m = .5$ and $\beta h = .54 > .5$, so the condition fails. Nevertheless $v_M^{\text{safe}} = .55$, $v_U^A(\ell) = .505$, and $\beta^2 h = .486$; hence every majority assessment strictly exceeds the endpoint unanimity vector $(.505, .505)$.

The excluded fraction is

$$\frac{e}{m} = \begin{cases} 1/2, & m \text{ even}, \\ (m-1)/(2m), & m \text{ odd}. \end{cases}$$

In an existing low-prior cell, the weaker local condition $\beta\ell < e/m$ gives the margin $\beta(e/m - \beta\ell)$. At $p = 0$, that condition guarantees only the ex ante conclusion unless the full counterfactual type vector is used.

E.6 Public-majority PBE forms

For public $o \leq 1/m$, each weak vote costs

$$r_M(o) = \frac{\beta(1-\beta o)}{m}.$$

Every optimal proposal pays exactly $r_M(o)$ to exactly k weak states, pays zero to the rest, and leaves

$$v_M^A(o) = 1 - \frac{k\beta(1-\beta o)}{m}$$

to H. Every probability measure over these minimal coalitions is a PBE proposal lottery. Passage strictly dominates the delayed payoff $\beta^2 o$.

For $o > 1/m$, every purchased weak vote costs β/m, so passage leaves v_M^{safe} and rejection yields βo. Thus:

$$\begin{aligned} 1/m < o < o_M^* &\;:\; \text{every PBE passes through a minimal coalition,} \\ o = o_M^* &\;:\; \text{every linked mixture of minimal passage and delay is allowed,} \\ o > o_M^* &\;:\; \text{every PBE deliberately delays.} \end{aligned}$$

At the tie, the same mixture weight binds the proposal, result, and agreement probability. The public payoff is always the singleton

$$v_M^A(o) = \begin{cases} 1 - \dfrac{k\beta(1-\beta o)}{m}, & o \leq 1/m, \\ \max\{v_M^{\text{safe}}, \beta o\}, & o > 1/m. \end{cases}$$

E.7 Public-unanimity PBE

Every weak vote costs

$$r_U(o) = \frac{\beta(1-\beta o)}{m}.$$

The unique optimal on-path proposal pays this amount to all m weak states and leaves

$$v_U^A(o) = 1 - \beta + \beta^2 o$$

to H. Rejection yields $\beta^2 o$, so passage is strictly better by $1-\beta$. Agreement is immediate and payoff-determined for every public type; no majority-style delay is imported into this rule.

E.8 The public gap

Subtracting E.6 from E.7 gives the unanimity-minus-majority gap

$$\Delta v^A(o) = \begin{cases} -\beta(e/m)(1-\beta o), & o \le 1/m, \\ \beta(\beta o - e/m), & o > 1/m,\ \beta o \le v_M^{\text{safe}}, \\ (1-\beta)(1-\beta o), & o > 1/m,\ \beta o \ge v_M^{\text{safe}}. \end{cases}$$

Continuity holds at $\beta o = v_M^{\text{safe}}$. For $o > 1/m$,

$$\operatorname{sgn} \Delta v^A(o) = \operatorname{sgn}(\beta o - e/m).$$

At $o = 1/m$, the first branch applies because the specified no-agenda continuation includes H at equality. This special selection also matters for the public agenda difference in E.12.

E.9 Exact informational-rent translations

For every complete private assessment,

$$IR_g^A = V_g^A - (v_g^A(\ell), v_g^A(h)).$$

This fixed-vector translation preserves every atom, mixture, and linked type coordinate. The exact unanimity cells are:

Table 9: Unanimity informational rent with hegemonic agenda

Parameter cell	IR_U^A	Ex ante image
$p=0$	$(0, \max\{v_U^A(\ell), \beta^2 h\} - v_U^A(h))$	0
$0<p\le p^*$, $v_U^A(\ell)\ge \beta^2 h$, $\mu^{\text{off}}=0$	$(0, -(v_U^A(h)-v_U^A(\ell)))$	$-p(v_U^A(h)-v_U^A(\ell))$
Other low-prior parameter cells	$\emptyset$	$\emptyset$
$p^*<p<1$, $\mu^{\text{off}}=0$	$\{(u - v_U^A(\ell), u - v_U^A(h)) : u \in [\max\{v_U^A(\ell), \beta^2 h\}, v_U^A(h)]\}$	$[\max\{v_U^A(\ell), \beta^2 h\} - v_{U,\text{ea}}^A(p), v_U^A(h) - v_{U,\text{ea}}^A(p)]$
$p^*<p<1$, $\mu^{\text{off}}\in(p^*,1]$	$(v_U^A(h)-v_U^A(\ell), 0)$	$(1-p)(v_U^A(h)-v_U^A(\ell))$
$p^*<p<1$, $\mu^{\text{off}}\in(0,p^*]$	$\emptyset$	$\emptyset$
$p=1$	$(v_U^A(h)-v_U^A(\ell), 0)$	0

Here $v_{U,\text{ea}}^A(p) = (1-p)v_U^A(\ell) + pv_U^A(h)$. Every existing cell satisfies

$$IR_U^A(\ell) \ge 0, \qquad IR_U^A(h) \le 0,$$

with at least one strict inequality. Majority rent is retained as the exact translation of the full majority payoff correspondence; it has no imposed global sign.

E.10 Public power and information decomposition

For every assessment in the fixed-specification comparison set,

$$\Delta IR^A(o) = IR_U^A(o) - IR_M^A(o) = \Delta V^A(o) - \Delta v^A(o),$$

or equivalently

$$\Delta V^A(o) = \Delta v^A(o) + \Delta IR^A(o).$$

If $\Delta v^A(\ell) < 0$ and a private assessment has $\Delta V^A(\ell) > 0$, then

$$\Delta IR^A(\ell) > -\Delta v^A(\ell) > 0$$

for that assessment. Type incidence precedes ex ante aggregation:

$$\Delta V^A_{\text{ea}} = \Delta v^A_{\text{ea}} + \Delta IR^A_{\text{ea}}, \quad \Delta v^A_{\text{ea}} = (1-p)\Delta v^A(\ell) + p\Delta v^A(h).$$

Under the sufficient condition in E.5,

$$\Delta V^A(o) \leq -\beta(e/m - \beta h), \qquad \Delta IR^A(o) \leq -\Delta v^A(o) - \beta(e/m - \beta h).$$

This bounds the informational difference relative to the amount required to reverse majority's guaranteed private advantage; it does not assign a sign to $\Delta IR^A(o)$.

The numerical assessment in Table 6 gives

$$\Delta IR^A(\ell) = .5480, \qquad \Delta v^A(\ell) = -.4095, \qquad \Delta V^A(\ell) = .1385.$$

These linked values illustrate the identity within the stated theoretical parameterization.

E.11 Agenda accounting identity

At date A, the four quantities being compared are

	B	A
public type	$\beta v^B_g(o)$	$v^A_g(o)$
private type	$\beta V^B_g(o)$	$V^A_g(o)$.

Define

$$\begin{aligned} D_g(o) &= v^A_g(o) - \beta v^B_g(o), \\ T_g(o) &= V^A_g(o) - \beta V^B_g(o), \\ I_g(o) &= IR^A_g(o) - \beta IR^B_g(o). \end{aligned}$$

Substituting $V = v + IR$ in the agenda and baseline games yields, by type and ex ante,

$$T_g = D_g + I_g.$$

The factor β appears once because the no-agenda game begins one date later. Thus T_g combines the earlier date with the mandatory proposal as specified by the agenda extension.

E.12 Public agenda differences across institutions

The rule-specific public differences are

$$D_U(o) = 1 - \beta$$

and

$$D_M(o) = \begin{cases} v_M^{\text{safe}} - (e/m)\beta^2 o, & o \le 1/m, \\ v_M^{\text{safe}} - \beta o, & 1/m < o < o_M^*, \\ 0, & o \ge o_M^*. \end{cases}$$

The threshold satisfies $o_M^* > 1/m$, but it can be weakly above one, in which case the last branch is empty in the admissible domain. At $o = 1/m$, the first branch applies. The specified continuation selection creates the jump

$$D_M(1/m) - \lim_{o \downarrow 1/m,\ o > 1/m} D_M(o) = \frac{\beta}{m}\left(1 - \frac{e\beta}{m}\right) > 0.$$

With $\Delta D(o) = D_U(o) - D_M(o)$,

$$\Delta D(o) = \begin{cases} -\beta(e/m)(1 - \beta o), & o \le 1/m, \\ \beta(o - e/m), & 1/m < o \le o_M^*, \\ 1 - \beta, & o \ge o_M^*. \end{cases}$$

The last two formulas coincide at o_M^*. These three branches give the complete classification supported by the model.

E.13 Total private agenda differences

Under unanimity, the exact linked correspondence is

$$T_U = \begin{cases} (1-\beta, \max\{v_U^A(\ell) - \beta^2 h, 0\}), & p = 0, \\ \emptyset, & 0 < p \le p^*, \\ \{(u - \beta^2 h, u - \beta^2 h) : u \in [\max\{v_U^A(\ell), \beta^2 h\}, v_U^A(h)]\}, & p^* < p < 1,\ \mu^{\text{off}} = 0, \\ \emptyset, & p^* < p < 1,\ \mu^{\text{off}} \in (0, p^*], \\ (1-\beta, 1-\beta), & p^* < p < 1,\ \mu^{\text{off}} \in (p^*, 1], \\ (1-\beta, 1-\beta), & p = 1. \end{cases}$$

In the low-prior interior, the no-agenda correspondence is empty; the agenda correspondence may be nonempty or empty. In the high-prior, low-positive off-path parameter cell, the baseline correspondence is nonempty and the agenda correspondence is empty. In both cases $T_U = \emptyset$.

Where both games have an assessment, every unanimity assessment weakly benefits both types and the maximum gain is $1 - \beta$. At $p = 0$, the ex ante gain is $1 - \beta$ even if the probability-zero high coordinate is zero.

For majority, retain the exact interaction correspondence:

$$T_M = D_M + I_M.$$

For each coordinate and assessment,

$$\begin{aligned} T_M(o) > 0 &\iff I_M(o) > -D_M(o), \\ T_M(o) = 0 &\iff I_M(o) = -D_M(o), \\ T_M(o) < 0 &\iff I_M(o) < -D_M(o). \end{aligned}$$

When the source correspondence crosses the threshold, its sign is selection-dependent. Across rules,

$$\Delta T(o) = \Delta D(o) + \Delta I(o)$$

assessment by assessment in every complete product. A robust ranking requires the entire admissible correspondence to lie on one side of $-\Delta D(o)$.

E.14 The diagonal contrast

The diagonal comparison is

$$Q_g(o) = v_g^A(o) - \beta V_g^B(o) = D_g(o) - \beta IR_g^B(o).$$

It compares public information with agenda against private information without agenda and therefore changes two model features at once.

Under unanimity,

$$Q_U = \begin{cases} (1-\beta, 1-\beta), & p = 0, \\ \emptyset, & 0 < p \leq p^*, \\ (v_U^A(\ell) - \beta^2 h, 1-\beta), & p^* < p \leq 1. \end{cases}$$

The ex ante value in the high-prior region is

$$Q_{U,\text{ea}} = 1 - \beta - (1-p)\beta^2(h-\ell).$$

Under majority,

$$Q_M = D_M - \beta IR_M^B$$

is the exact image of the applicable linked no-agenda assessments and remains set-valued where those assessments do. Q_U may exist where T_U is empty because it uses the public agenda game and the private baseline game.

Appendix F: Factorization, interaction, and scope

F.1 Exact and economic layers

For majority, the exact signature is

$$\mathrm{Sig}^{ex}_M(R_M) = (\rho, \mu^{\mathrm{off}}, \Lambda_{(\Gamma^{M,R_M}_\ell, \Gamma^{M,R_M}_h)}),$$

where Λ codes the diagonal relabeling orbit of the two complete realized laws. Its economic summary is

$$\mathrm{Sum}^{econ}_M(R_M) = (\rho, \mu^{\mathrm{off}}, (q_M)_\# \Gamma^{M,R_M}_\ell, (q_M)_\# \Gamma^{M,R_M}_h).$$

Unanimity uses the analogous pair Sig^{ex}_U and Sum^{econ}_U. The exact signature preserves the diagonal relabeling orbit of the realized laws and the relevant identities inside those realized tuples. Complete off-path beliefs, strategies, supports, messages, and continuation functions remain in the underlying equilibrium assessment and are used for every off-path-sensitive operation. The economic layer retains the anonymous realized laws needed for payoffs, agreement, delay, and anonymous outcomes.

The institutional comparison is first applied to complete assessments. After that product is formed do payoff and outcome maps factor measurably through the economic summaries:

$$\mathcal{C}_{\text{econ}}(R_M, R_U) = \overline{\mathcal{C}}_{\text{econ}}(\mathrm{Sum}^{econ}_M(R_M), \mathrm{Sum}^{econ}_U(R_U)).$$

The orbit law indexes diagonal relabeling equivalence. Equality of economic summaries identifies the anonymous realized laws, while named proposals, coalition lotteries, and off-path plans remain assessment-level objects. Appendix B.9 proves that the exact orbit law is a complete invariant and that every anonymous Borel outcome used here factors uniquely through the economic summary.

For every fixed specification with a nonempty correspondence under both rules, let $V^A_{g,o}(R_g)$ denote H's date-A payoff of type $o \in \{\ell, h\}$ encoded by the complete assessment R_g, and define its ex ante coordinate by

$$V^A_{g,\text{ea}}(R_g) = (1-p)V^A_{g,\ell}(R_g) + pV^A_{g,h}(R_g).$$

For $s \in \{\ell, h, \text{ea}\}$, define the two scalar images and their exact contrast by

$$\begin{aligned} M_s &= \{V^A_{M,s}(R_M) : R_M \in \mathcal{B}_M(\mathbf{d}, \rho, \mu^{\text{off}})\}, \\ U_s &= \{V^A_{U,s}(R_U) : R_U \in \mathcal{B}_U(\mathbf{d}, \rho, \mu^{\text{off}})\}, \\ \mathcal{C}^A_s &= \{u - v : u \in U_s,\ v \in M_s\}. \end{aligned}$$

The contrast envelope satisfies

$$\inf \mathcal{C}^A_s = \inf U_s - \sup M_s, \qquad \sup \mathcal{C}^A_s = \sup U_s - \inf M_s.$$

The interval between these numbers is an interval hull, not necessarily an attained set. For $g \in \{M, U\}$ and $o \in \{\ell, h\}$, write

$$\bar{\Gamma}^{o,R_g}_g := (q_g)_{\#} \Gamma^{g,R_g}_o.$$

The outcome object is the ordered pair of marginal anonymous laws

$$\mathcal{O}_A(\mathbf{d}, \rho, \mu^{\text{off}}) = \left\{ ((\bar{\Gamma}^{\ell,R_M}_M, \bar{\Gamma}^{h,R_M}_M), (\bar{\Gamma}^{\ell,R_U}_U, \bar{\Gamma}^{h,R_U}_U)) : (R_M, R_U) \in \mathcal{J}^{\text{cmp}}_A(\mathbf{d}, \rho, \mu^{\text{off}}) \right\}.$$

This object contains the ordered marginal anonymous laws. A common cross-rule draw or coordinate splicing would require an additional model primitive.

F.2 Interaction with the no-agenda benchmark

Let IR^B_g be the native Round-1 rent vector from the no-agenda benchmark. The exact interaction is the Minkowski difference of complete linked assessments,

$$I_g = IR^A_g - \beta IR^B_g.$$

For majority, the applicable no-agenda assessments are the three public regions used in Proposition 5.6; no general sign is inferred.

Under unanimity:

$$I_U = \begin{cases} (0, \max\{v_U^A(\ell), \beta^2 h\} - v_U^A(h)), & p = 0, \\ \emptyset, & 0 < p \le p^*, \\ \left\{ \begin{array}{c} (u - v_U^A(h), u - v_U^A(h)) : \\ u \in [\max\{v_U^A(\ell), \beta^2 h\}, v_U^A(h)] \end{array} \right\}, & p^* < p < 1,\ \mu^{\text{off}} = 0, \\ (0,0), & {\scriptstyle p^*<p<1, \atop \mu^{\text{off}} \in (p^*,1]}, \\ \emptyset, & {\scriptstyle p^*<p<1, \atop \mu^{\text{off}} \in (0,p^*]}, \\ (0,0), & p = 1. \end{cases}$$

Thus agenda weakly reduces unanimity's informational rent in every existing high-prior parameter cell; the change is strict for both types exactly when $\mu^{\text{off}} = 0$ and $u < v_U^A(h)$. At $p = 0$, only the probability-zero high coordinate falls.

The institutional interaction is

$$\Delta I = \Delta IR^A - \beta \Delta IR^B.$$

It exists only when both exact sources exist. When either source is set-valued, the interaction preserves each source's linked payoff coordinates.

F.3 Payoff dates, existence, and verification limits

All agenda-game values arrive at date A. Native no-agenda Round-1 values are multiplied by β exactly once before comparison. Public agenda payoffs, private agenda payoffs, agenda rents, and their institutional contrasts receive no additional factor.

The existence rules are:

- every public agenda game has a PBE for both rules and types;
- IR_g^A exists exactly where the private agenda source exists;
- ΔIR^A exists exactly where the fixed-specification comparison set exists;
- I_g exists exactly where both its agenda and no-agenda rent sources exist;

- ΔI, T_g, and ΔT exist only where every source required by their definitions exists.

An absent source propagates $\emptyset$ through every definition that uses it. Appendices B.7–B.9 prove the completeness and existence claims for the agenda correspondences, the absence of a third unanimity family, and the measurability and factorization properties used in the comparison.

F.4 Interpretive scope of the agenda comparison

The quantity T_g compares two specified games: the extension inserts an earlier, mandatory hegemonic proposal stage, while the baseline begins in Round 1. Consequently, T_g includes the value of timing built into those games. Its substantive interpretation is limited to this model-defined change, rather than an optional proposal right or an empirical estimate for the WTO.

When a contrast is set-valued, each comparison is assessment-specific. A selection-free claim requires the entire exact correspondence to lie on the relevant side of the threshold. The stated results are the source correspondences on their declared domains.

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